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Multi-resolution enhancement for full-spectrum neural representations

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Scientific data acquisition continues to outpace storage and analysis capabilities, making voxel-based representations increasingly intractable. Implicit neural representations (INRs) offer a promising solution by encoding signals through coordinate-based neural networks, serving as surrogates of data, with computational and storage requirements scaling with network complexity rather than data dimensionality. However, smaller INRs struggle to faithfully represent multiscale structures, high-frequency information and fine textures that constitute a large proportion of scientific measurements. We propose WIEN-INR, a theoretically guided hierarchical INR framework that distributes modelling across resolution scales and enables improved representation capacity through a novel enhancement network to recover subtle details. This multiscale architecture allows smaller networks to retain the full spatial-frequency content of the signal as well as preserve training efficiency and lower storage cost. Evaluated on distinct raw experimental measurements across scales and complexities, WIEN-INR represents a practical step towards a broader adoption of neural representations in scientific workflows, delivering compact, robust and high-fidelity representations.

The rise and continued development of implicit neural representations (INRs) have gained traction and momentum from computer vision and graphics into scientific task flows—from experimental steering[1], data analysis and visualization[2–5] to data compression/transmission[6,7]. At the same time, as scientific facilities generate increasingly massive measurements at accelerating rates, the analysis and storage of the resulting voxel-based datasets are becoming more unsustainable, creating an urgent need for a compact and scalable data framework[8–10].

Rather than treating data as discrete snapshots on voxel grids, INRs view it as samples from an underlying continuous and structured field. This perspective aligns with the observation that most scientific data do not occupy the entire ambient space, but rather lie on a lower-dimensional embedding. Hence, an INR can usually provide a more compact representation than storing values on a full coordinate grid. INRs learn a generator function that maps the input coordinates (for example, physical parameter space) to the corresponding signal values (for example, intensities), encoding the information in the parametric weights of the network[11]. It offers several distinctive advantages for scientific data pipelines, including the following: memory efficiency that scales with model complexity rather than resolution[6,12,13]; it is modality agnostic, which avoids the generalization issues typical of dataset-trained deep encoding/decoding methods; it offers continuous functional fitting and differentiability with gradient access; and resolution independence and region-of-interest (ROI) decoding through coordinate-based queries[14,15].

However, the representational capacity of conventional INRs—designed primarily for natural images, videos and scenes—is generally less reliable when modelling complex scientific data and can struggle

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to capture physics-relevant information. Prior scientific applications typically adopt or build on off-the-shelf INRs as task components; our experiments show that these conventional architectures can underfit the physics-related characteristics intrinsic to scientific measurements (Fig. 1), especially with smaller networks. Moreover, using an INR as a storage alternative requires a high-fidelity, near-one-to-one representation of the underlying signal. An ideal scientific INR framework should simultaneously (1) preserve the full spectrum of physics-related information, (2) generalize robustly across modalities and (3) be computationally and storage efficient, with fast encoding and low-latency decoding.

These goals suggest the need for careful consideration in network design and theoretical guidance. Compact networks are favourable for fast encoding, but are often biased towards low-frequency components that tend to capture smooth components before high-frequency (HF) details[14,16], whereas larger networks offer enhanced capacity to capture finer structure, but are often costly to train and store. More broadly, identifying which features can be removed without degrading science-relevant information is non-trivial: what appears as 'noise' for a given experiment may be a signal for another[17]. In the ideal case, increasing the network size guarantees universal approximation[18], meaning that sufficiently large models can represent any measurable function arbitrarily well. In practice, however, this theoretical capacity faces constraints. First, existing INR architectures are often not parameter efficient, with capacity not aligned towards task-relevant structures. Second, simply enlarging the network does not always ensure convergence to the desired accuracy, particularly for fine-scale or HF structures, which are often more difficult to capture accurately. These limitations highlight the need for compact, full-spectrum representations (that is, representations that retain the full range of spatial frequencies) that preserve both global structure and localized detail, allowing small networks to achieve high fidelity without loss of efficiency.

Here we present the wavelet integrated enhancement network (WIEN-INR), an INR-based hierarchical framework that operates in the multi-resolution wavelet domain. WIEN-INR deploys multiple subnetworks to model different time–frequency contents and uses a compact enhancement module to increase the representation capacity of small coordinate-based INRs to better capture HF details. The proposed architecture is supported by a mathematical theory; it is designed with the objective of retaining full-spectrum information and achieving shorter training time and smaller networks for storage. We tested WIEN-INR across a diverse set of scientific datasets, including high-resolution X-ray diffraction[19], four-dimensional (4D) inelastic neutron scattering[20], ultrafast X-ray scattering[21], coherent diffraction imaging[22], ptychography[23] and astrophysical data[24], demonstrating notable improvements in rate distortion, spectral fidelity, texture preservation and robustness across domains of vastly different scales and modalities. We also evaluate the sensitivity of the method to different hyperparameter choices and network configurations, demonstrating robust performance that supports generalization with minimal tuning costs.

Results

Our approach is motivated by two key observations. First, achieving representations that faithfully preserve the scientifically relevant details and remain compact requires architectures that can effectively capture meaningful information across the full frequency spectrum. Second, compact networks must be endowed with mechanisms that enhance their representational capacity. As illustrated in Fig. 1, existing INR frameworks, when constrained to a small parameter budget, often over-smooth fine details or introduce unphysical textures, as exemplified by a representative raw X-ray speckle dataset. We demonstrate that WIEN-INR achieves accurate representations that preserve fine-scale details and textures that are physically meaningful and maintain a compressive network size, delivering strong rate-distortion performance.

WIEN-INR

The design of WIEN-INR is guided by two principles. The first is that the rate distortion can be improved by operating in a domain that better decouples and decorrelates information across scales. We achieve this by applying a wavelet transform to partition the signal into different frequency bands across scales (a detailed discussion of wavelet over Fourier is provided in the Methods). The second principle is that compact coordinate-based multilayer perceptrons (MLPs) can be endowed with additional representational power through a lightweight enhancement module, enabling them to capture subtle details without inflating the network size. Both components are modular: the wavelet transform acts as a preprocessing step, and the enhancement module can be integrated with any coordinate-based INR.

Architecture. Our approach addresses the limitations of using small INRs to represent the full-spectrum information, particularly HF details. The wavelet transform decomposes the input data into frequency bands that are better decorrelated, helping alleviate the low-frequency bias in compressive INRs by separating HF details into dedicated sub-bands. We further improve the model efficiency by using a shared network per scale frequency band, taking advantage of the correlation between wavelet coefficients[25,26]. For the band with the finest details, which is particularly challenging for small INRs to represent, we introduce a novel enhancement module that increases the representational capacity without substantially increasing the model size. This module infers finer-scale details conditioned on coarser-scale estimations, effectively acting as a learnable local operator that enhances the ability of the network to capture the finest details.

Positioning to state-of-the-art INR compressors. Existing INR-based compression methods pursue various strategies to improve the encoding speed and storage efficiency. COIN[27] and COIN++[28] fit MLPs to the target signal and then quantize the model weights; COIN++ further improves the rate and encoding speeds by learning a meta-learned base network that captures shared structure and encoding each instance via lightweight modulations. Strümpler et al.[12] combine sparsity/low-rank constraints with quantization-aware training and entropy coding. These approaches implicitly assume a sufficiently capable and stable base INR for the target data. By contrast, the design of WIEN-INR directly targets the fundamental limitation of small INRs—the difficulty of capturing full-spectrum information under parameter budget.

Theory. The design choices are further supported by our theoretical analysis of split INRs in the lazy regime or the neural tangent kernel (NTK) regime[29,30] (a formal definition is given in Supplementary Section C3). We treat the NTK as the design variable induced by an INR of bounded size and consider the HF band error. For a wavelet band indexed by (j, i), let $\mathcal{H}_{j,i}$ denote the (fixed) band subspace and $P_{j,i}$ denote the associated projection operator. Let $P \in \mathbb{N}$ be a scalar budget equal to the total number of trainable parameters (weights and biases). Define

$$\mathcal{K}^{\text{all}}(P) := \{K \mid K \text{ is the NTK of some INR that outputs all wavelet coefficients with} \le P \text{ parameters}\}.$$

For the feasibility of block-diagonal constructions, define the class of realizable band-restricted kernels with parameter $\theta_{j,i}$ as

$$\mathcal{K}_{j,i}(p_{j,i}) := \{K_{j,i} \mid K_{j,i} \text{ is the NTK of a single head on band } (j,i), |\theta_{j,i}| \le p_{j,i}, \ \text{Ran}(K_{j,i}) \subseteq \mathcal{H}_{j,i}\}.$$

Any band-split INR with per-band budgets $\{p_{j,i}\}$ realizes NTK $K_{\text{blk}} := \bigoplus_{(j,i)} K_{j,i} \in \mathcal{K}^{\text{all}}\left(\sum p_{j,i}\right)$ (see Lemma 1 in the Methods).

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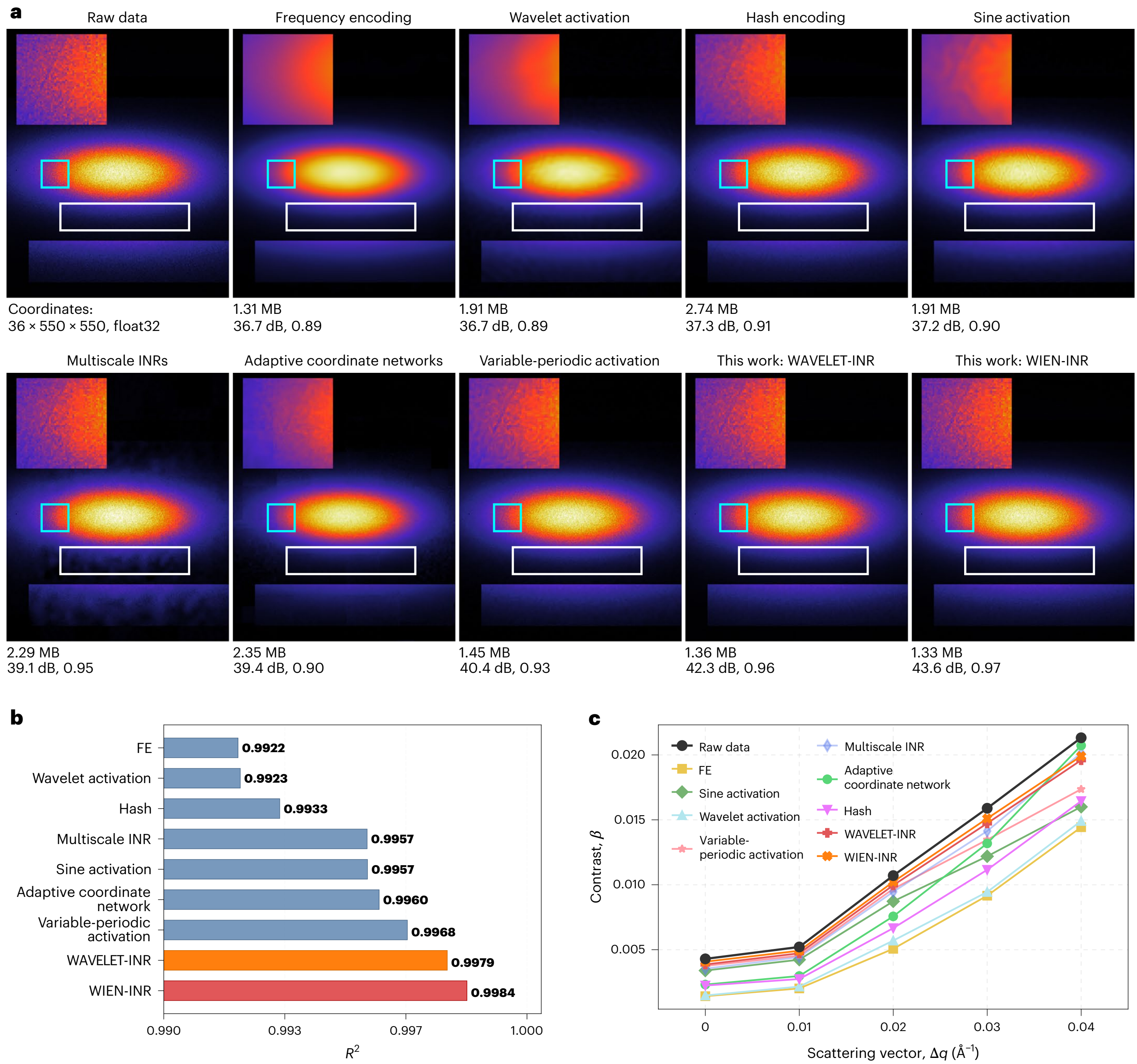

Fig. 1 | Efficient INRs preserve HF and physics-related details. Comparison of INRs of raw volumetric data (36 × 550 × 550, float32) collected from a high-resolution wide-angle coherent X-ray scattering experiment from the intermetallic alloy Cu_3Au (ref. 19). Each trained network serves as a compressed surrogate for the original volume: instead of storing the raw voxel array, the information is encoded in the network weights. **a**, Representative slice decoded by querying the trained INRs at the corresponding spatial coordinates and rendering the network outputs as pixel intensities. In all cases, the network size is smaller than the raw data representation, with parameter counts labelled beneath each panel. A unique challenge for coherent X-ray scattering datasets such as this is that fine-structure speckle constitutes the signal of interest, whereas in many other imaging modalities, this is treated as noise. Insets: zoomed-in regions of high-speckle intensity, highlighting the superior ability of our method to capture fine details and textures. Competing INRs either over-smooth the data (frequency encoding[16], wavelet activation[33], hash encoding[13], sine activation[14] and adaptive coordinate networks[36]), lose speckle contrast or fail to preserve weak diffuse scattering (variable-periodic activation[34] and multiscale INRs[35]). Our methods preserve the speckle contrast and achieve the highest PSNR and SSIM with fewer parameters counts, with metrics reported below each panel. **b**, Beyond producing visually faithful representations, our method also achieves the highest R^2 on the full volume, indicating pixel-wise consistency between the decoded and ground-truth signals. **c**, An important metric for the scattering data is speckle contrast, defined as the statistical variance of intensities divided by their mean. The contrast varies with momentum transfer q, that is, a mapping that changes with distance from the peak centre. We plot the q-dependent contrast as the ROI is translated horizontally. Our method preserves the trend with 95% contrast retained. Because the network is inherently compressive (with WIEN-INR ~30 times smaller than the raw data), further increasing its size enables closer matching of the trends.

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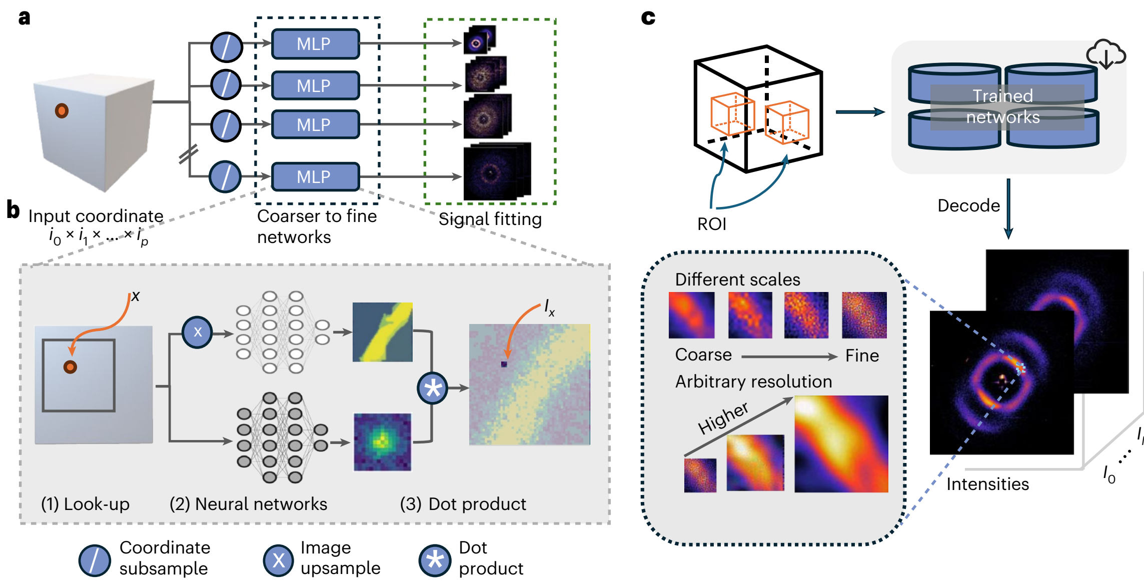

Fig. 2 | Schematic of the WIEN-INR architecture. a, We design a hierarchical encoding process. Encoding begins with the input of p-dimensional coordinates (for example, detector coordinates $\mathbf{x} = [i, j]$), which are normalized and passed to separate MLPs, each trained to predict the wavelet coefficients (coarse to fine) of the signal at different resolution scales. **b**, Within each MLP, we introduce a novel enhancement module that refines the output of the previous coarser-scale MLP by (1) extracting a local neighbourhood of size $(2n_r + 1)$ per input dimension, centred at the input coordinate $\mathbf{x}$ in coefficients space; (2) inputting the extracted local neighbourhood to two INRs—one uses coordinates upsampled from the neighbourhood to query the coarser-scale MLP to provide an approximation of the higher resolution (this network is fixed), and the other, a learnable INR that acts as local refinement; (3) and then applying the dot product to the outputs of these two networks to match the target finer-scale wavelet coefficients. This operation is rasterized across all coordinates of interest, refining the coarser outputs into finer-scale outputs. **c**, After training, the decoding process allows users to query specific ROIs at any desired resolution using the outputs of the separate networks. To map the outputs to the true signal domain, the localized IDWT is applied to the output coefficients. Beyond the improved rate distortion and accuracy, the hierarchical encoding and decoding process also supports multiscale analysis (from coarse to fine details), revealing phenomena at different frequency bands. Beyond scale, the INR-based framework supports decoding at an arbitrary resolution: without interpolation, one can query increasingly fine resolutions to obtain a truly continuous representation.

Let f_0 the initial prediction and $\mathbf{y}$ represent the true coefficients. Fix $t > 0$, initial residual $r_0 = f_0 - \mathbf{y}$ and an HF error threshold $\varepsilon_{\mathrm{HF}}$. Let $P_{\mathrm{sep}}(\varepsilon_{\mathrm{HF}})$ be the minimal parameters for a band-split INR to achieve HF error $\varepsilon^2_{\mathrm{HF}}$, and $P_{\mathrm{mono}}(\varepsilon_{\mathrm{HF}})$ denote the minimal parameters among all INRs (possibly coupled across bands):

$$P_{\mathrm{sep}}(\varepsilon_{\mathrm{HF}}) := \inf\Big\{P \geq 0 : \exists \text{ block-diagonal } K \in \mathcal{K}^{\mathrm{all}}(P) \text{ s.t. } \textstyle\sum_{(j,i)\in \mathrm{HF}} \| e^{-tK} P_{j,i} r_0 \|^2_{L^2} \leq \varepsilon^2_{\mathrm{HF}} \Big\},$$

$$P_{\mathrm{mono}}(\varepsilon_{\mathrm{HF}}) := \inf\Big\{P \geq 0 : \exists K \in \mathcal{K}^{\mathrm{all}}(P) \text{ s.t. } \sum_{(j,i)\in \mathrm{HF}} \| e^{-tK} P_{j,i} r_0 \|^2_{L^2} \leq \varepsilon^2_{\mathrm{HF}} \Big\}.$$

Main theorem 1. Split is capacity-optimal at a fixed HF target: *let $P_{\mathrm{sep}}(\varepsilon_{\mathrm{HF}})$ and $P_{\mathrm{mono}}(\varepsilon_{\mathrm{HF}})$ be defined as above. Under NTK linearization and Lemma 1,*

$$P_{\mathrm{sep}}(\varepsilon_{\mathrm{HF}}) = P_{\mathrm{opt}}(\varepsilon_{\mathrm{HF}}) \leq P_{\mathrm{mono}}(\varepsilon_{\mathrm{HF}}).$$

Equality holds if and only if the monolithic model's realized NTK at optimum is block-diagonal over $\bigoplus_{(j,i)\in \mathrm{HF}} \mathcal{H}_{j,i}$ and induces the same optimal band allocation.

Remark 1. (Training-time and spectral bias)

The above result uses only parameter-budget constraints and holds for any $t > 0$. It does not exploit or assume frequency-dependent contraction rates. In particular, we do not assume (nor rule out) inequalities such as $\lambda_{\min}(K|_{\mathcal{H}_{\mathrm{LF}}}) \gg \lambda_{\min}(K|_{\mathcal{H}_{\mathrm{HF}}})$ (precise definitions are provided in Supplementary Section C4c) that encode spectral bias. Empirically, the gradient descent fits the low-frequency content faster than HF signals (spectral bias[31]), and periodic activation INRs (for example, SIREN) mitigate[14] but do not eliminate this effect, as observed in our experiments. Consequently, under a small training-time budget, the observed gap between split and monolithic designs can be larger than our parameter-only analytic result. A rigorous theory that jointly accounts for parameter budgets, spectral bias and finite-time optimization is an interesting direction for future work.

On the basis of the above theoretical justification, we propose a new architecture, with a schematic shown in Fig. 2. The network is trained by minimizing the ℓ_2 loss between the network's outputs and the wavelet coefficients at each wavelet scale (equation (1)). A novel enhancement module is introduced to further improve the representation power of small networks towards HF information. During training, we fix a target compression level and allocate the total parameter budget across scales according to the intrinsic data dimensionality: coarser scales are assigned smaller subnetworks, whereas finer scales receive larger subnetworks to better capture details.

We evaluate WIEN-INR on distinct scientific datasets, including wide-angle coherent X-ray scattering[19], inelastic neutron scattering[20], ultrafast XFEL X-ray scattering[21], coherent diffraction imaging[32], ptychography[23] and the astrophysical Helioseismic and Magnetic Imager[24],

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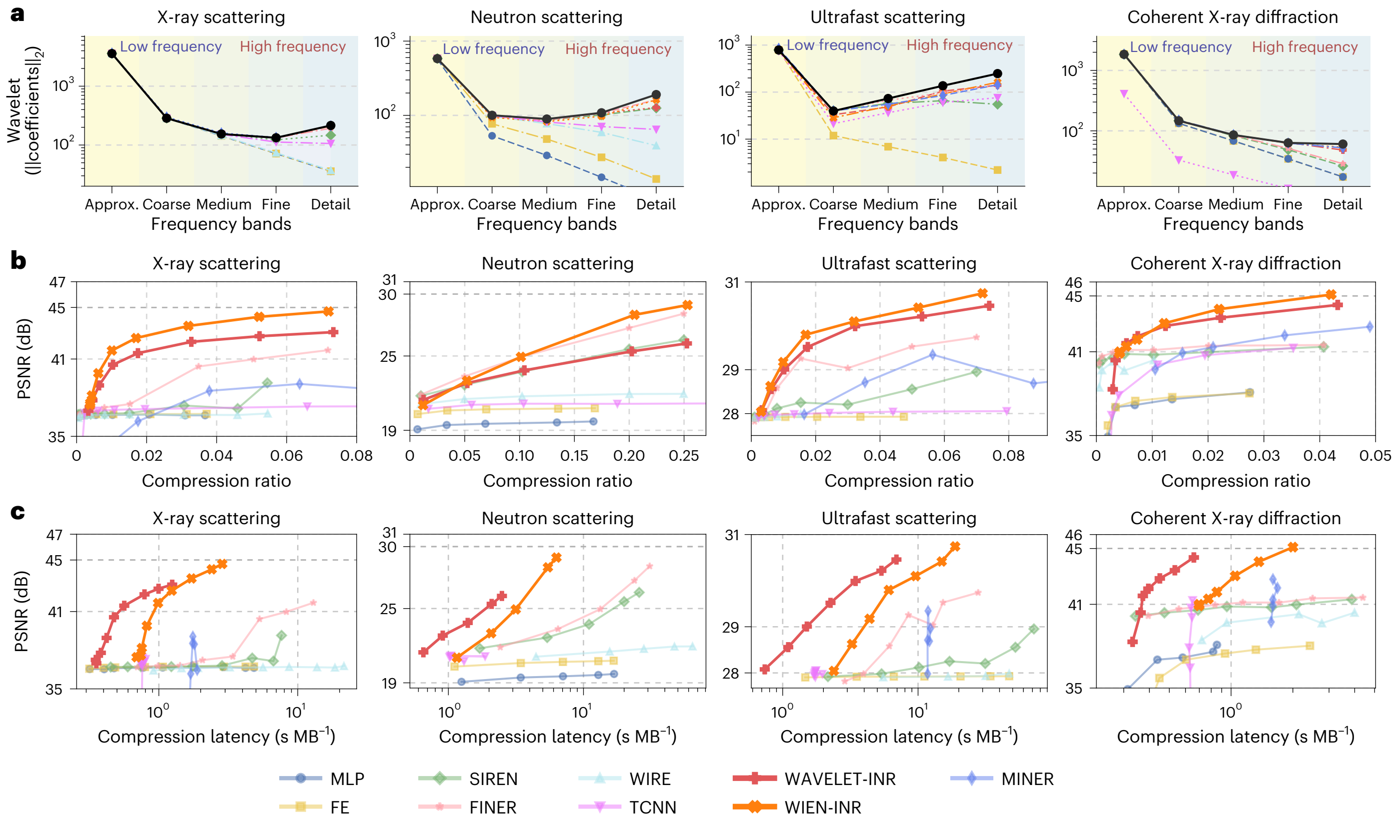

Fig. 3 | Improved rate distortion. a, An important metric to quantify whether representations preserve the full spectrum of information is how well the decoded wavelet coefficients match those of the original signal. The wavelet transform is known to be superior to Fourier analysis in transient or localized phenomena, as different scales capture more complete information: coarser bands represent broad, low-frequency trends, whereas finer bands capture localized phenomena, such as sharp transitions and HF details[25]. We quantify fidelity across scales by comparing the ℓ_2 norm of the wavelet coefficients between the decoded and original signals. Results are shown across four scales and the approximation coefficients of the reconstructed DWT (Haar), sub-bands for four benchmark scientific datasets. Although all baselines remain close to the ground truth at coarse levels, they deviate for finer scales, WIEN-INR closely follows the true DWT distribution (black line), demonstrating robust recovery of fine-scale content and full-spectrum fidelity. **b**, Rate-distortion comparison across four scientific datasets, showing compression rate versus PSNR, with stronger methods achieving higher accuracy at smaller compression rates. WIEN-INR consistently attains a higher PSNR than competing INR methods at the same compression rates (that is, with fewer network parameters). Performance gains depend on the data characteristics, with the greatest improvements on datasets containing dense, fine-scale structure (for example, the X-ray measurements), where the adaptive time–frequency allocation and finest band enhancement of WIEN-INR is the most effective. **c**, Another important aspect of the utility is encoding efficiency, that is, the time to achieve a given accuracy level. We plot the encoding latency versus the accuracy achieved (PSNR) for the same networks used in **a** and **b**, trained for the same number of global epochs. WAVELET-INR achieves faster per-epoch encoding due to simpler architecture, but reaches a lower maximum PSNR. MINER scales up nicely to larger models due to its multiscale, multipatch design and block selection/dropping[35]. Please note that MINER was not implemented on the neutron scattering data due to limitations posed by the 4D input dimension.

each probing different physical properties. A detailed description of the datasets, including their physical origins and experimental objectives, is provided in Supplementary Section A.

For comparison, we benchmark INRs designed with frequency awareness or multi-resolution considerations. We cover different families of methodologies, including sinusoidal activations (SIREN)[14], wavelet activations (WIRE)[33] and variable-periodic activations (FINER)[34] that tune the network's spectral bias to better represent higher-frequency content; input encodings such as Fourier feature encoding (FE)[16], which lifts coordinates into a sinusoidal feature space, and multi-resolution hash encoding (TCNN)[13], which uses a multi-resolution learned feature grid for fast fitting; additionally, multiscale INRs such as MINER[35], which use patch-wise MLPs in a Laplacian-pyramid refinement that benefits from spatially sparse signals, and adaptive coordinate networks (ACORN)[36], which learn an adaptive quadtree/octree-style partition of the spatial domain to allocate resources based on local signal complexity. We also include a standard rectified linear unit MLP baseline and the proposed WAVELET-INR baseline (equation (3)).

Improved rate distortion

We quantify the accuracy using the peak signal-to-noise ratio (PSNR), the structural similarity index (SSIM) and pixel-wise correlation. The compression rate is defined as

$$\text{Compression rate} = \frac{Pb_{\text{param}}}{Nb_{\text{data}}},$$

where P is the number of network parameters; N is the dimensionality of the original tensor; and b_{param} and b_{data} are the bits per parameter or data element, respectively.

Figure 3 presents our main numerical results, showing that our method consistently achieves improved rate-distortion performance compared with all benchmarks. The results reveal three critical advantages: preservation of full-spectrum information (Fig. 3a), higher accuracy at comparable compression ratios (Fig. 3b) and higher accuracies with comparable encoding times (Fig. 3c). The enhancement module further differentiates WIEN-INR from the baseline WAVELET-INR approach by endowing networks of the same size with higher representation power. It provides better initialization as it transitions between

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wavelet scales and enables improved convergence to bands with the finest details (Extended Data Fig. 1). Additionally, we also evaluate traditional compression methods using the traditional codec JPEG2000 (ref. 37), autoencoder-based neural compression BMS[38] and scientific compression codec SZ3 (refs. 39–41) (Extended Data Figs. 2–4); the implementation details are provided in Supplementary Section B2.

Preservation of physics-related features

The ability to preserve fine-scale features and signals of varying orders of intensity is particularly crucial, where both subtle details and weak signals often contain valuable physical information. We evaluate fidelity using standard global metrics such as PSNR, SSIM and R^2, and, where possible, complement them with local and physics-related analyses to identify where errors occur and whether they affect scientifically important structures. As shown in Fig. 1, WIEN-INR accurately reproduces the speckle patterns in the X-ray scattering data of the binary alloy Cu_3Au, preserving both the intensity distribution (Fig. 1b) and contrast (Fig. 1c) that are critical for subsequent physical analysis—despite using a network more than 30 times smaller than the raw data size.

The qualitative results across multiple scientific datasets measured at XFEL, neutron and synchrotron sources demonstrate the modality-agnostic ability of WIEN-INR to preserve critical features in scattering measurements and maintain a compact representation. In these experiments, speckle constitutes the signal of interest, whereas in many other imaging fields, it is treated as noise and removed. Additionally, the preservation of weak signals is crucial in scattering measurements, as physically important features often appear in low-intensity regions. Consistent with its stronger global fidelity, WIEN-INR also achieves ~2–7 times smaller local mean squared error across Bragg peaks and weak-scattering ROIs, whereas the baseline methods (SIREN, FINER and MINER) produce various visual artefacts and unphysical textures (Fig. 4 and Extended Data Fig. 5). Complementary decoded volumes under more constrained model-size budgets are provided in Extended Data Fig. 6. Additionally, multiscale methods often benefit from data with spatial sparsity or where signal content is unevenly distributed across space—typical of natural signals. By contrast, speckle-dominated scattering measurements contain dense, low-amplitude but information-bearing structures across the full field of view and span multiple scales, making purely spatial-complexity-based allocation or block pruning less effective (Supplementary Fig. 6).

For reconstruction tasks, we also evaluated WIEN-INR on compressing raw coherent diffraction imaging[22] and coherent X-ray imaging measurements[23]. This setting is non-trivial because the encoded raw measurements must be subsequently processed with numerical inverse solvers to obtain image-space reconstructions. We assess quality after decoding and reconstruction with the standard downstream pipeline and, as shown in Fig. 5, reconstructions from WIEN-INR maintain improved fidelity compared with those obtained using SIREN.

Data characteristics and performance

WIEN-INR's multiscale decomposition and finest-scale enhancement provide more gains for datasets with rich fine-scale structure. Speckle-dominated X-ray and XFEL scattering data show the largest improvements in both encoding speed and reconstruction quality (Fig. 3 and Extended Data Fig. 5). Beyond scattering, experiments on solar magnetogram data demonstrate similar gains for complex spatiotemporal structures (Extended Data Fig. 7). At the other end of the spectrum, for encoding smaller and smoother natural images, the average PSNR gains are modest—and can even decrease at low bit rates due to multiscale overhead (Extended Data Fig. 8). In particular, improved visual qualities near sharp boundaries and high textures are still evident.

Faster encoding with decoding latency

Encode–decode efficiency is another practical bottleneck for deploying INRs in scientific workflows that require fast analysis, as INRs can be slow to encode. Splitting training across wavelet scales, WIEN-INR yields smaller, downsampled learning problems per band (~2^{pj} in p-dimensional at scale j, as derived in Supplementary Section C1). Further acceleration is also possible through parallelization of the independently trained bands, since the loss is essentially separable. Table 1 reports the time to encode with WIEN-INR and the best-performing benchmark methods, across scales of network sizes that are trained for fixed durations. In particular, given the equivalent training time with comparable memory requirements, WIEN-INR consistently achieves the highest accuracy (PSNR). The decoding times of WIEN-INR are reported in Table 2. The decoding process involves the evaluation of neural networks at relevant scales for specific ROIs, followed by the inverse discrete wavelet transform (IDWT). Detailed implementation of the localized ROI decoding for WIEN-INR is provided in Supplementary Algorithm 2.

Robustness to hyperparameters

WIEN-INR is robust to the underlying hyperparameters, including the choice of wavelet filters (for example, Haar and db4), the number of wavelet decomposition levels J and the refinement window size r for the enhancement module (Extended Data Fig. 9). A detailed discussion of the wavelet choices is provided in Supplementary Sections C1 and C2.

We further evaluated the robustness of WIEN-INR by varying the width and depth of the enhancement network. Supplementary Fig. 2 shows training curves across different network configurations. WIEN-INR (Supplementary Fig. 2) consistently converges faster to a higher PSNR with smoother transitions into the band with the finest details compared with the WAVELET-INR baseline (Supplementary Fig. 3). To isolate the contribution of the upsampled initialization, we tested a residual baseline that learns the difference between upsampled coarse outputs and finest-scale targets (Supplementary Section B4). This residual approach performs no better than directly training a SIREN on the finest band (Supplementary Fig. 4), indicating that the enhancement module's convolutional design is essential for the improved representation and convergence on the challenging speckle-dominated data with dense fine-scale features.

Discussion

Our method demonstrated reliable performance in achieving more efficient and accurate scientific data representation with a compressive network. Meanwhile, we identify several promising directions for further improvement.

Network compression and weight quantization

In our experiments, we applied a simple post-training FP32-to-FP16 quantization of network weights to reduce memory cost. More sophisticated approaches from the model compression literature could be explored in future work, including the meta-learned network[28], deep compression pipeline and quantization-aware training methods that optimize for rate distortion under low-precision arithmetic[6,42,43]. Beyond quantization, structured pruning and sparsification techniques may provide further reductions in model size[44]. We demonstrated a possible pipeline for post-training compression in Supplementary Section B5.

Further speed-up

As the loss is separable across scales, each subnetwork can, in principle, be trained independently. The only exception is at the finest scale at which the enhancement network depends on the output of the network trained on the previous finest scale. This structure allows for a parallelized implementation, which would drastically reduce the overall training time, with the runtime effectively dominated by the training of the next finest scale and the enhancement step. Additionally, in the context of INR-based representations in which encoding is traditionally slow, recent advances such as divide-and-conquer strategies,

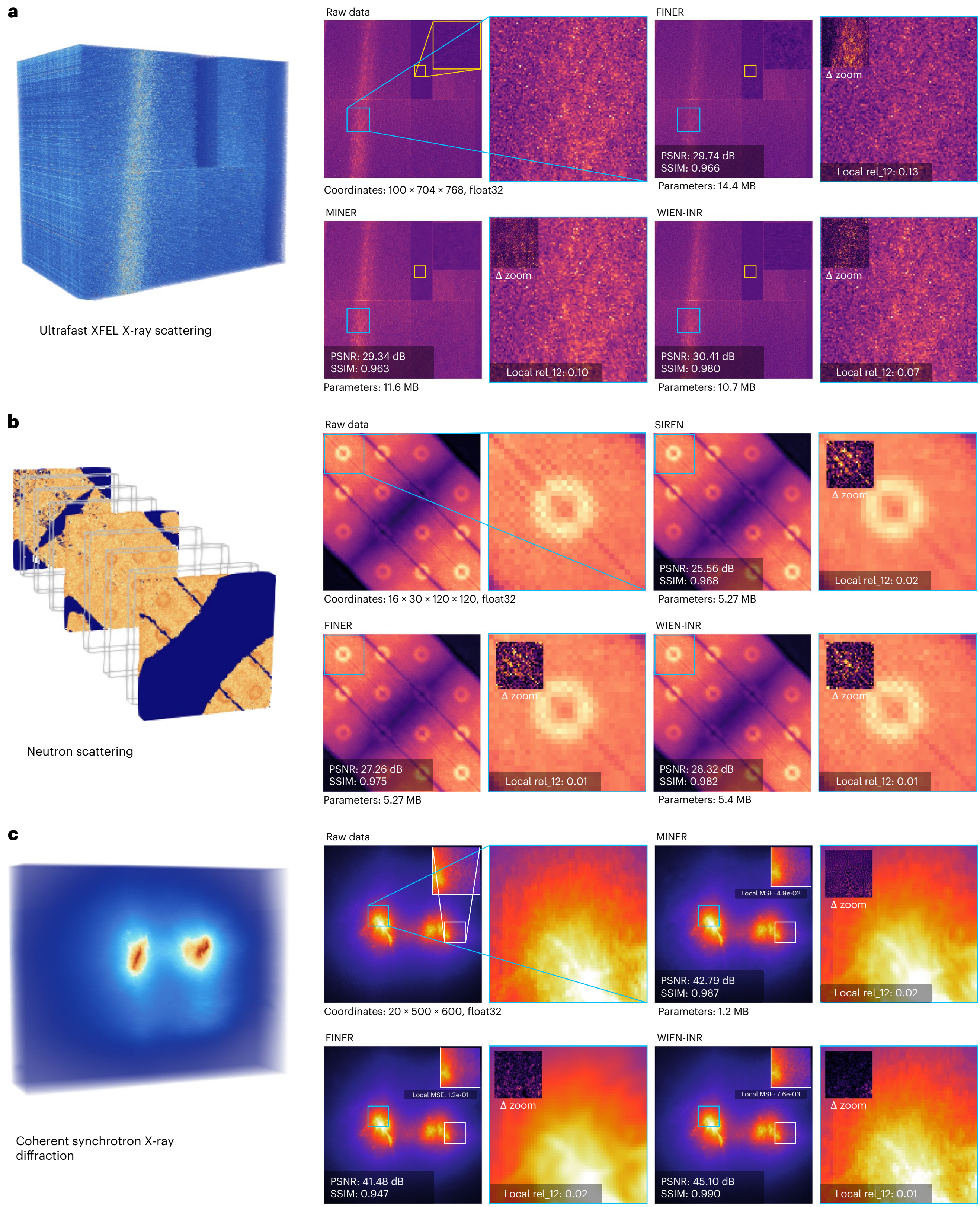

Fig. 4 | Robustness of WIEN-INR across datasets measured at different physics experiments. a–c, Reconstruction results are shown for three distinct datasets collected at XFEL (**a**), neutron (**b**) and synchrotron (**c**) facilities[20,21], corresponding to the ultrafast scattering, neutron scattering and coherent X-ray diffraction columns from Fig. 3. Each volume is encoded and decoded to visualize if the network can focus training on physically important features. The representative two-dimensional visualizations of the decoded three-dimensional/4D volumes with zoomed-in views are shown on the right. The PSNR and SSIM metrics of the full volume are reported in the inset, whereas the zoomed-in views also display local loss metrics. Across modalities, these comparisons highlight that SIREN, FINER and MINER[14,34,35]—the strongest baselines—exhibit lower PSNR and higher local relative ℓ_2 error, with reconstructions that either lose speckle contrast (over-smoothing) or create unphysical features (patches or unphysical textures). In all cases, WIEN-INR remains robust both visually and quantitatively.

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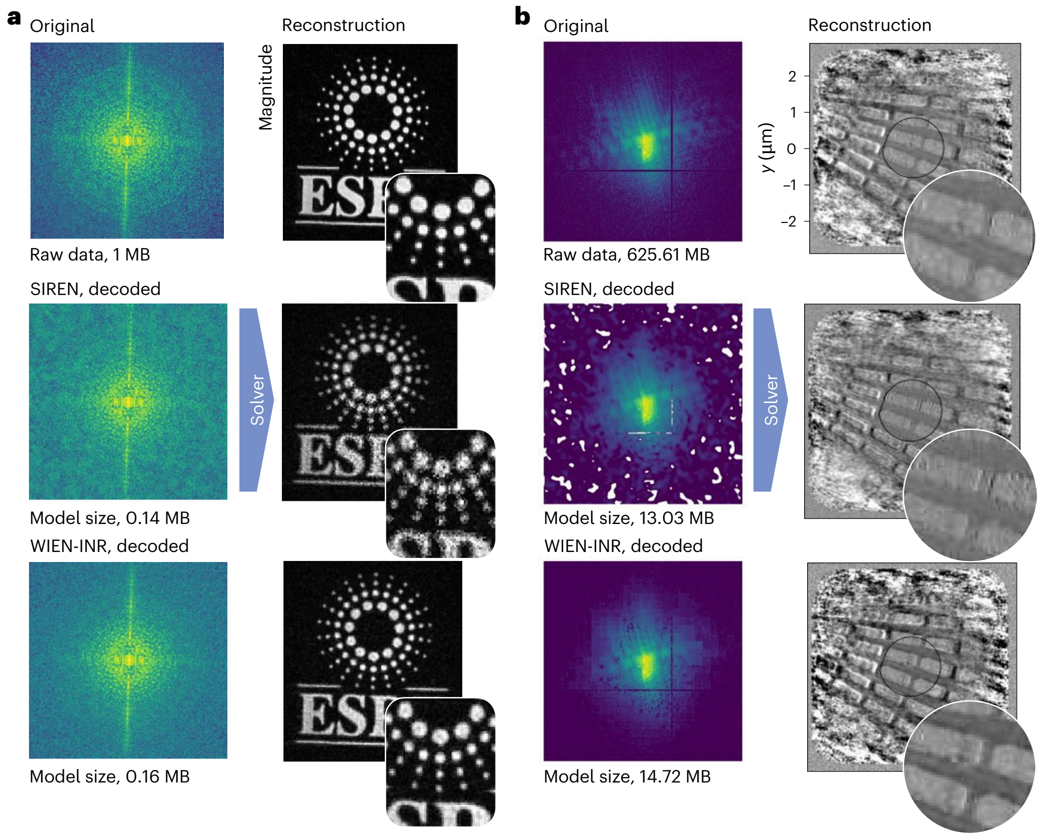

Fig. 5 | Reconstruction quality using the decoded signal. For tasks in which information is revealed through subsequent reconstruction, it is crucial that the decoded signal minimally distorts reconstruction. This task is highly non-trivial because during encoding/training, losses are typically minimized only in the measurement space, which does not directly translate into reconstruction fidelity. **a**, A use case of reconstructing an ESRF logo from INR-encoded and INR-decoded coherent diffraction data[22], using a combination of error reduction and hybrid input–output algorithm. We plot the magnitude of the reconstruction space: the top row shows the reconstruction using the original measurements without neural compression; the middle row shows reconstruction using the best-performing baseline network (SIREN, 0.14 MB); and the bottom row shows reconstruction using WIEN-INR (0.16 MB). Visual degradation is evident in the SIREN case, whereas WIEN-INR retains a higher edge contrast. **b**, Reconstruction of a ptychography measurement[23] (625 MB) at a higher compression rate using SIREN (13.03 MB) and WIEN-INR (14.72 MB). Both networks encode and decode the full volume of diffraction measurements, followed by reconstruction through joint optimization of the probe and object using the alternating projection algorithm. A representative slice of the decoded diffraction (plotted in the log scale) is shown in the left column, with the reconstruction shown in the second column. In the central region of the reconstructed object, SIREN exhibits clear smearing, whereas WIEN-INR preserves sharp details.

Table 1 | Encoding latency

Dataset	X-ray scattering (Cu_3Au)			Neutron scattering			X-ray scattering (ultrafast)		
Network size	Small	Medium	Large	Small	Medium	Large	Small	Medium	Large
PSNR (dB)									
WIEN-INR: <1 min	37.45 (30 s)	42.62 (52 s)	42.90 (56 s)	23.00 (55 s)	21.86* (58 s)	21.49* (60 s)	28.51 (57 s)	28.70 (53 s)	28.48 (52 s)
WIEN-INR: <2 min	37.60	42.62	44.69 (119 s)	23.00	24.93 (83 s)	26.64* (119 s)	28.04 (99 s)	29.52 (119 s)	29.71 (111 s)
WIEN-INR: <7 min	37.60	42.62	44.69	23.00	24.93	29.66 (188 s)	28.63 (135 s)	29.80 (253 s)	30.10 (6 m 38 s)
Others: <1 min	36.71|SIREN	37.12|TCNN	37.45|TCNN	21.79|SIREN	21.06|TCNN	26.91|MINER	27.90|TCNN	27.99|TCNN	28.03|TCNN
Others: <2 min	36.89|TCNN	37.23|FINER	39.07|MINER	21.87|FINER	24.44|MINER	28.73|MINER	27.98|TCNN	28.02|TCNN	28.06|TCNN
Others: <7 min	36.89|TCNN	37.51|FINER	40.98|FINER	23.35|FINER	24.95|FINER	28.73|MINER	27.98|SIREN	29.25|FINER	28.06|TCNN

INRs are generally slow at encoding. For comparable network sizes (small, medium and large), we evaluate the accuracy after 1, 2 and 7 min of training on the Cu_3Au sample[19], neutron scattering[20] and ultrafast scattering[21], respectively. Entries marked with an asterisk correspond to intermediate WIEN-INR results that had not reached the finest-band optimization stage within the specified time budget. Across all three benchmark datasets and the completed runs, WIEN-INR outperforms the competing methods in most model/time configurations: for each comparable network size, it achieves the highest PSNR within the same training time. This demonstrates its utility when encoding time is critical.

Table 2 | Decoding latency

Dataset	X-ray scattering (Cu_3Au)				
Network size	Parameters (MB)	PSNR (dB)	Median (ms)	p90 (ms)	Throughput ($MVox\,s^{-1}$)
WIEN-INR (small)	0.42	41.7	12.83	12.88	848.75
WIEN-INR (medium)	0.7	42.6	13.68	14.04	795.85
WIEN-INR (large)	1.33	43.6	14.23	14.44	765.37

The decoding process only requires network evaluation at inference over the ROI. We report the decoding time for reconstructing the full 41.5-MB volume using WIEN-INR with a small (0.42 MB), medium (0.7 MB) and large (1.33 MB) network configuration. The reported metrics include median latency (ms), 90th-percentile latency (ms) and throughput ($MVox\,s^{-1}$).

meta learning and block-wise training[45,46] could further be adopted to accelerate training, making computationally efficient INR frameworks increasingly practical.

Advanced training strategies

Advanced training strategies, such as batch normalization[47], gradient adjustment methods (inductive gradient adjustment INR[48]) and more recent rank-preserving optimization[49] or initialization strategies[50], may further improve HF learning by reshaping optimization dynamics. Empirical comparisons with batch normalization and inductive gradient adjustment INR are shown in Supplementary Fig. 7. Inspired by the adaptive spatial allocation of MINER/ACORN[35,36], patch-wise training could further improve encoding: although patch-wise refinement may introduce discontinuities at the finest scale, coarser scales could benefit from localized fitting and selective refinement. This approach is reminiscent of JPEG2000 (ref. 51), which applies wavelet transforms and then encodes sub-bands using localized code blocks for efficient spatial coding.

Learnable transformations

Alternatively to handcrafted wavelet transforms, one could learn the transform itself in a data-driven way, turning the framework into a dictionary learning approach. Promising directions include parameterizing the lifting filters with small networks under perfect-reconstruction constraints, training shallow autoencoders to discover orthogonal wavelet bases, or making scattering/wavelet banks partially learnable to adapt to the data[52,53]. Finally, one can also explore parameterized analytic wavelet families or even architectures that bypass a fixed transform altogether. For example, Kolmogorov–Arnold networks[54] are a flexible, data-adaptive multiscale transform that could bypass the need for a fixed basis while retaining interpretability.

Generative framework

Since the current method is designed for a faithful, compressive representation of data, another promising extension is into the generative setting. In particular, the coarse-to-fine refinement naturally enables super-resolution by predicting fine-scale details from coarser representations, which we leave for future work. Additionally, wavelet-domain diffusion models have shown how coarse-to-fine decomposition can guide generative synthesis of three-dimensional shapes[55], whereas generative neural fields based on mixtures of implicit functions enable the efficient sampling of new signals[56]. In a similar spirit, our multi-resolution plus enhancement structure could be adapted to learn priors over coarse INR representations and enhancement kernels, enabling the synthesis of high-fidelity scientific data.

Conclusion

We presented WIEN-INR for scientific data representation, designed to retain full-spectrum information and achieve competitive rate-distortion performance. Our contributions are as follows. (1) Multi-resolution decomposition and fine-scale enhancement: by leveraging wavelet transforms and the enhancement module, the method naturally separates frequency content, enabling the efficient learning of more spectrally controlled data representations that are often lost in conventional INRs. (2) Mathematical proof establishing our multiscale architecture is optimally efficient for representing HF information under the NTK scheme. (3) Flexible architecture: each wavelet band is represented by an independent subnetwork, with enhancement modules applied selectively to scales at which naive INRs are insufficient. This provides both scalability (wavelets can be computed in patches), parallelism (subnetworks can be trained independently) and control (users may choose how many bands to preserve). Additionally, the per-pixel accuracy can be set by choosing an appropriate error bound tolerance through various loss functions such as mean absolute error, mean squared error, SSIM or perceptual losses. This multi-resolution framework is well-suited to encode complex data when full-spectrum information must be retained under a limited parameter budget. After encoding, the representation is a queryable, differentiable neural field that natively supports downstream tasks, for example, ROI-based decoding, denoising, parameter fitting, registration or downstream inference. The decoding stage is nearly instantaneous—requiring only forward neural network evaluations. By explicitly addressing the fundamental limitations of conventional INRs in representing complex scientific data, we expect WIEN-INR to broaden the utility of neural representations across data-intensive scientific domains.

Methods

Overview of INRs

The central idea behind INR is to convert explicit information into implicit model weights through unsupervised training. Specifically, given an input coordinate $\boldsymbol{x} \in \mathbb{R}^p$, we are interested in learning a function that maps $\mathbf{x}$ to a quantity of interest $\boldsymbol{y} \in \mathbb{R}^d$. Such a mapping can be approximated by a neural network, typically an MLP, $\varphi_\omega(\mathbf{x})$ parameterized by the weights ω:

$$\varphi_\omega : \Omega \subset \mathbb{R}^p \longrightarrow \mathbb{R}^d, \quad \mathbf{x} \mapsto \varphi_\omega(\mathbf{x}),$$

where p is the input dimension and d is the output dimension (for example, $d = 1$ for scalar fields and $d = 3$ for RGB values).

To query a multidimensional tensor $\boldsymbol{I} \in \mathbb{R}^{D_1 \times D_2 \times \ldots \times D_p}$, we evaluate the network on a coordinate grid. Here

$$\mathcal{I} = \{(i_1 \ldots i_p) : 1 \le i_k \le D_k\}$$

denotes the full set of index tuples. Each index $(i_1, \ldots i_p) \in \mathcal{I}$ corresponds to a normalized coordinate $\mathbf{x}_{i_1,\ldots,i_p} \in [-1,1]^p$ and a tensor value $\mathbf{I}[i_1, \ldots i_p] \in \mathbb{R}^d$.

An INR is trained to minimize the discrepancy between its predictions and the ground-truth tensor values at all discrete coordinates:

$$\min_{\omega} \sum_{(i_1,\ldots i_p)\in\mathcal{I}} \mathcal{L}\left(\varphi_\omega(\boldsymbol{x}_{i_1,\ldots,i_p}), \boldsymbol{I}[i_1, \ldots i_p]\right), \tag{1}$$

where $\mathcal{L}$ denotes a suitable loss function. In practice, this full-sum objective is optimized over mini-batches by sampling from $\mathcal{I}$. The training objective in equation (1) is conceptually straightforward, yet convergence is not guaranteed by the objective alone[14,57].

Overview of wavelet transformations and multi-resolution analysis

Inspired by wavelet theory, we explore a hierarchical INR design that encodes the wavelet coefficients of the data instead of working with the original space. In principle, a sufficiently expressive coordinate MLP could emulate multiscale transforms (and even surpass them) by virtue of universal approximation. In practice, however,

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optimization dynamics make this inefficient: standard INRs exhibit a well-documented spectral bias—they fit low frequencies first and struggle to recover the HF structure; therefore, convergence on fine detail is slow or requires substantially larger models[14,16].

The INR community has developed many notable techniques to mitigate the inherent spectral bias of the network. A few approaches include positional encoding[16,34] (which maps input coordinates to higher-dimensional features), hash encoding[13] (which introduced an efficient multi-resolution grids of trainable features for compact representation) and alternative activation functions such as sinusoidal or wavelet activations (for example, SIREN[14] and WIRE[33]). Additionally, more recent methods focus on improving model architectures to better capture fine details and a wider range of frequencies, such as INCODE, FR and MFN[58–60].

The question of identifying which features can be safely removed without affecting the representation/reconstruction quality is highly non-trivial, and what seems like 'noise' to a model may be a signal to a researcher[17]. A partial answer lies in selecting an appropriate analysis window—such as a block size or time interval that ensures meaningful patterns in the data are effectively captured[25]. This choice involves a trade-off: smaller windows allow for the detection of localized, HF phenomena or 'anomalies', whereas larger windows tend to smooth over these details and are better suited for identifying broader, low-frequency trends[25]. From signal processing, we know that methods that rely on the assumptions of stationarity or ergodicity over large windows (for example, Fourier analysis) tend to obscure transient or localized phenomena, particularly when the underlying data are random or non-stationary[25,26].

Wavelet theory and multiscale analysis offer a compelling solution to this trade-off by enabling the simultaneous examination of both local anomalies and global trends. In the wavelet domain, the coefficients provide a hierarchical description of the signal: some capture long-range correlations with narrow frequency bands, whereas others represent short-term variations associated with wide frequency bands. This multiscale representation inherently allows for a flexible balance between resolution in time (or space) against resolution in frequency (or momentum). An illustration of the wavelet transform coefficients and its time–frequency tiling is shown in Extended Data Fig. 10. Additional details on the technical details of wavelet transform are provided in Supplementary Section C1.

This multiscale capability is particularly desirable in physics data analysis in which the phenomena of interest often manifest across a range of spatial or temporal scales.

WIEN-INR

While preparing this paper, Yu et al.[61] independently proposed CF-INR, which also directly encodes the wavelet coefficients for INR learning. Yu et al. use a single-level Haar decomposition with self-evolving parameters and leverage a low-rank structure in the wavelet domain, which is well-motivated for natural images but can be less suitable for speckle-dominated scientific measurements. An empirical comparison is provided in Supplementary Fig. 8. By contrast, WIEN-INR adopts a multiscale wavelet decomposition with a dedicated coarse-to-fine enhancement module targeting the finest details, designed for compact, high-fidelity scientific data representation.

A multi-resolution preprocess: the first step in WIEN-INR is to apply a wavelet transform to the input tensor $\mathbf{I}$, decomposing it into frequency bands that are better decorrelated. The time–frequency localization property of the wavelet transform helps alleviate the low-frequency bias in compressive INRs by isolating HF details into dedicated sub-bands. We denote the wavelet coefficients of general p-dimensional data with approximation coefficients denoted by $\mathbf{a}_J$, and detail coefficients of the jth detail scale with direction i as $\mathbf{d}_j^i, 1 \le j \le J, 1 \le i \le 2^p - 1$. A straightforward approach to operate in the wavelet space is to apply a discrete wavelet transform (DWT) to the signal and then train separate small INRs to model each sub-band independently, and finally reconstruct the signal using the IDWT. The training process is as follows:

$$\min_{\omega_j, \bar{\omega}_J} \sum_{j=1}^{J} \sum_{i=1}^{2^p-1} \left\| \varphi_{\omega_j}^i - \mathbf{d}_j^i \right\|^2 + \left\| \phi_{\bar{\omega}_J} - \mathbf{a}_J \right\|^2. \quad (2)$$

Each neural network $\varphi_{\omega_j}^i, \phi_{\omega_J} : \Omega_j \mapsto \mathbb{R}^d$ (we write $\varphi_{\omega_j}^i, \phi_{\bar{\omega}_J}$ as φ_j^i, ϕ_J from now on) is typically parameterized as an MLP that maps spatial coordinates $\mathbf{x} \in \Omega_j$ at the appropriate resolution to the corresponding wavelet coefficient.

However, this naive formulation suffers from the following drawbacks: (1) although wavelets decorrelate sub-bands in a second-order sense, the formulation ignores the strong statistical dependencies across scales and orientations, which are well-exploited in wavelet theory and zero-tree coding[25,26]; (2) generic INR architectures struggle to converge for the finest scale (that is, $\mathbf{d}_j^i$) due to their limited representational capacity when constrained to small network sizes. These noted shortcomings call for a more tailored architecture to handle the compression of the finest band and, ultimately, to produce improved-quality encoded data.

Model correlation with a shared INR per scale: induced by the geometric image regularity[25], the large-magnitude coefficients in $\mathbf{d}_j^i$ tend to be aggregated along contours or in textured regions. Indeed, wavelet coefficients have a large amplitude at which the signal has sharp transitions. At each scale j and for each orientation i, a wavelet image coder can take advantage of the correlation between neighbouring wavelet coefficient amplitudes, referred to as intraband correlation. Note that taking advantage of this intrascale amplitude correlation is an important source of improvement for traditional compression techniques, such as in JPEG2000 compression[51].

Taking this into account, instead of using independent INRs for each of the $2^p - 1$ directions of the value i, we can use a single INR with an output dimension of $2^p - 1$ such that it encodes information from all directions simultaneously. Specifically, we train a single φ_{ω_j} for the scale j such that $\varphi_{\omega_j} : \mathbf{x} \to \mathbb{R}^{d(2^p-1)}$. Furthermore, the optimization problem becomes

$$\min_{\omega_j, \bar{\omega}_J} \sum_{j=1}^{J} \left\| \varphi_{\omega_j} - \left(\mathbf{d}_j^1, \cdots, \mathbf{d}_j^{2^p-1} \right) \right\|^2 + \left\| \phi_{\bar{\omega}_J} - \mathbf{a}_J \right\|^2 \quad (3)$$

Ideally, by adjusting the frequency bias of each φ_{ω_i}—for example, via the frequency scaling parameter in SIREN or frequency encoding—we can align each network with the frequency range for scale j it is intended to model (for example, the time–frequency tile in Extended Data Fig. 10). We call this new formulation in equation (3) as WAVELET-INR, which serves as the foundation for our enhanced architecture.

Enhancement module: from our experiments, we observe that the band with the finest details (approximately corresponding to frequency $\omega \in [\pi/2, \pi]$) is particularly challenging for small INRs to represent (Supplementary Fig. 3). Although one possible remedy might be to further decompose this band (for example, using wavelet packets[52]), we argue that the difficulty does not stem primarily from frequency bias. Instead, it arises because generic INRs—especially in their compressive form—struggle to effectively capture and reproduce such fine-scale textures (Extended Data Fig. 1). To overcome this limitation, we introduce a tailored INR-based neural architecture that increases the representational capacity for any coordinate-based INRs without incurring a substantial increase in model size.

This is achieved by inferring finer-scale details conditioned on coarser-scale representations: for each fixed orientation i, suppose we can approximate scale j with a lightweight INR $\varphi_j^i : \Omega_j \to \mathbb{R}^d$. Then, for the next finer scale $(j-1)$ of the same i, the coefficient can be approximated by super-resolving φ_j^i onto the next scale coordinates Ω_{j-1}:

$$\varphi_{j-1}^i : \Omega_{j-1} \to \mathbb{R}^d, \text{ such that } \varphi_{j-1}^i(\mathbf{x}) = \varphi_j^i(\mathbf{x}) \text{ for } \mathbf{x} \in \Omega_j \subset \Omega_{j-1}. \quad (4)$$

This upsampling operation is possible due to the continuity of the INR, and can be done recursively up to the finest detail Ω_1. However, these upsampled networks do not contain higher-frequency information beyond the scale on which they were trained. Instead, they provide an approximate prediction of the next scale, serving primarily as a geometric before guiding the encoding of finer details—similar in spirit to zero-tree coding[26].

To super-resolve the fine details, we introduce a lightweight predictor network using MLP that acts as a local learnable operator. Specifically, for a fixed orientation i, we train a predictor $P^i_{j\to j-1} : \Omega_{j-1} \to \mathbb{R}^{dr^p}$ to super-resolve the coarser upsampled predictions. It takes as input a coordinate $\mathbf{x} \in \Omega_{j-1}$ and outputs a kernel of size r for each input dimension (that is, r^p in p dimensions). Conceptually, this kernel corresponds to a local receptive field centred at $\mathbf{x}$. During training, for each coordinate $\mathbf{x}$ in Ω_{j-1}, we do the following. (1) Extract an r^p patch from the upsampled coarser-scale INR prediction φ^i_j, centred at $\mathbf{x}$. (2) Apply the kernel produced by the predictor $P^i_{j\to j-1}(\mathbf{x})$ to this patch (via the inner product). (3) Match the resulting scalar prediction to the true wavelet coefficient $\mathbf{d}^i_{j-1}(\mathbf{x})$. The training objective, therefore, is

$$\min_{\xi} \sum_{i,\mathbf{x}\in\Omega_{j-1}} \left\| \mathbf{d}^i_{j-1} - \left\langle \varphi^i_{j-1}(\mathcal{N}_r(\mathbf{x})), P^i_{j\to j-1,\xi}(\mathbf{x}) \right\rangle \right\|^2, \quad (5)$$

where $\mathcal{N}_r(\mathbf{x})$ denotes the r^p neighbourhood of $\mathbf{x}$ in the upsampled coarse prediction. The complete formulation described above constitutes our proposed WIEN-INR framework.

This idea is reminiscent of a progressive refinement in which a network learns to predict the distortion between a coarse approximation and the true signal[3]. Note that the enhancement module is stand-alone and can be applied to any coordinate-based INR. The enhancement module is essential to the success of HF detail preservation. We also tested a more conceptually straightforward alternative enhancement approach that uses a residual network to model the difference $\{\mathbf{d}^i_{j-1} - \varphi^i_{j-1}\}_i$; however, we observed that this approach does not work better than simply using a larger network to learn $\mathbf{d}^i_{j-1}$ only. Details are provided in Supplementary Section B4.

HF capacity with NTK block structure

For the theoretical results, we work on the hypercube $\Omega = [0, 1]^p$ and assume the network is trained in the lazy/NTK regime[29,30], that is, wide SIREN/MLP architectures with NTK parameterization and small updates linearize around initialization so that the training dynamics are governed by a fixed NTK. In our setting, the INR does not regress the signal $\mathbf{I}$ itself; rather, it directly regresses the wavelet coefficient fields of $\mathbf{I}$. We note that one can derive the same conclusion for the more general NTK or gradient flow setting by working with spectral bias towards a low-frequency signal. However, we defer a detailed analytic study to further work.

Wavelet bands and HF error in coefficient space: fix an orthonormal, compactly supported wavelet basis $\{\psi_{j,i,k}\}$ on $L^2(\Omega)$ adapted to the boundary[25]. For each scale $j \in \{1...J\}$ and orientation $i \in \{1...2^p - 1\}$, let

$$\mathcal{H}_{j,i} := \operatorname{span}\{\psi_{j,i,k} : k \in \mathbb{Z}^p\} \subset L^2(\Omega), P_{j,i} : L^2(\Omega) \to \mathcal{H}_{j,i}$$

be the associated band subspace and orthogonal projector. Let $\mathrm{HF} \subseteq \{(j,i)\}$ denote the set of HF bands (for example, all $j \geq J_0$). Writing $\mathbf{I} = \sum_{j,i,k} d_{j,i}(k)\psi_{j,i,k}$, the INR with the band details is trained to predict the detail coefficient fields $\mathbf{y} = \{\mathbf{d}_1 ... \mathbf{d}_J\} = \{d_{j,i}\}_{(j,i)}$ bandwise. Because the wavelet transform is an orthonormal isometry, squared error in coefficient space equals squared error in signal space. We, therefore, measure the HF error of a predictor f (which outputs the coefficient fields) by

$$E_{\mathrm{HF}}(f) := \sum_{(j,i)\in\mathrm{HF}} \| P_{j,i}(f - \mathbf{y}) \|^2_{L^2},$$

where $P_{j,i}(f - \mathbf{y})$ means reconstruct the bandwise residual via the inverse wavelet transform restricted to $\mathcal{H}_{j,i}$ (equivalently, take the L^2 norm of the coefficient residual on band (j, i)).

NTK gradient flow for coefficient learning: for wide INRs and squared loss, gradient flow on parameters equals kernel gradient flow with a fixed NTK K value[29,30]. For each time t, writing $r_t := f_t - y$ (in coefficient space, equivalently in L^2 via the wavelet isometry), the dynamics satisfy

$$\partial_t r_t = -K[r_t], \ \| P_{j,i} r_t \|^2_{L^2} \leq e^{-2\lambda_{\min}(K|_{\mathcal{H}_{j,i}})t} \ \| P_{j,i} r_0 \|^2_{L^2},$$

where $\lambda_{\min}(K|_{\mathcal{H}_{j,i}})$ is the smallest eigenvalue of K restricted to $\mathcal{H}_{j,i}$.

Lemma 1. (Additivity and bandwise block structure) *Let the INR decompose as a sum of disjoint heads $f = \sum_{(j,i)} f^{(j,i)}$, each with its own parameter block. Then, the NTK is additive: $K = \sum_{(j,i)} K_{j,i}$. If, in addition, head (j, i) is trained only on the band-(j, i) coefficient target (equivalently, its effective output is $P_{j,i} f^{(j,i)}$), then $\operatorname{Ran}(K_{j,i}) \subseteq \mathcal{H}_{j,i}$ and $K = \bigoplus_{(j,i)} K_{j,i}$ is block-diagonal over the orthogonal decomposition $\bigoplus_{(j,i)} \mathcal{H}_{j,i}$.*

A complete proof is provided in Supplementary Section C4a.

Theorem 1. (Block-diagonal NTK is HF-optimal under the budget) *Let $\mathcal{K}(P) \equiv \mathcal{K}^{\mathrm{all}}(P)$. Then,*

$$\min_{K\in\mathcal{K}(P)} \sum_{(j,i)\in\mathrm{HF}} \| e^{-tK} P_{j,i} r_0 \|^2_{L^2} = \min_{\{K_{j,i}\in\mathcal{K}_{j,i}(p_{j,i}), \sum p_{j,i} \leq P\}} \sum_{(j,i)\in\mathrm{HF}} \| e^{-tK_{j,i}} P_{j,i} r_0 \|^2_{L^2}.$$

In particular, some optimizer is block-diagonal across bands: $K^\star = \bigoplus_{(j,i)\in\mathrm{HF}} K^\star_{j,i}$.

A complete proof is provided in Supplementary Section C4b.

Implementation details

For the subnetworks that represent each scale (that is, $\{\varphi_{\omega_j}, \phi_{\tilde{\omega}_j}\}$), we use independent SIREN networks with varying widths and depths, where the sine-layer frequencies are varied across scales but fixed across experiments, which controls the distribution of frequencies the network represents. The subnetwork size is controlled solely by adjusting the hidden-layer dimensions. The enhancement kernel network is also implemented using a SIREN, with a tunable hidden-layer width but fixed frequency parameters: $\omega_0 = 30$ for the first layer and $\omega = 30$ for the hidden layers. Details about the configurations for the Cu_3Au sample experiments are summarized in Supplementary Tables 3 and 4. SIREN is used as a backbone as it consistently achieved the best stand-alone performance across the scientific datasets examined in this work. We choose to apply the enhancement structure to the band with the finest details (that is, $\mathbf{d}_2$ level to the $\mathbf{d}_1$ level). This choice is data dependent (for example, the distribution of information across wavelet bands), but we found it sufficient for all evaluated scientific datasets in which the most challenging information to compress resides in the finest details. In practice, the enhancement network can be applied at any scale.

We implemented our framework on one NVIDIA A100 GPU with 40-GB memory. To optimize inference and decompression, all the trained model weights are stored in half-precision (2 bytes per entry). For training stability, we maintain a master copy of the parameters in full precision and use mixed-precision updates. A more detailed description of the network, training and hardware are provided in Supplementary Section B.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The primary Cu_3Au datasets that support the main claims and figures of this study are available via Zenodo at https://doi.org/10.5281/zenodo.17344185 (ref. 19). Neutron scattering data are available via

Zenodo at https://doi.org/10.5281/zenodo.8267499 (ref. 62). Additional publicly available datasets used in this study include the following: coherent diffraction imaging of the ESRF logo[32]; Siemens-star ptychography data from the PyNX ESRF public folder[63]; solar magnetogram data from the NASA Solar Dynamics Observatory Helioseismic and Magnetic Imager, analysing the first 20 h from 25 February 2014[24]; and the Kodak dataset from the Kodak True Color Image Suite[64]. The coherent X-ray scattering from Fe(Te,Se) and ultrafast X-ray scattering data from single-crystal $NiPS_3$ were collected under beamtime agreements that restrict public sharing but are available from J.J.T.

Code availability

The code for WIEN-INR, including scripts to regenerate the main results and baseline implementations, is publicly available via GitHub at https://github.com/YN754/INC4Sci.git. For reproducibility, the specific version used in this article is archived through a Code Ocean capsule (https://doi.org/10.24433/CO.8219927.v1)[65].

References

1. Chen, Z. et al. Implicit neural representations for experimental steering of advanced experiments. *Cell Rep. Phys. Sci.* **6**, 102333 (2025).
2. Chitturi, S. R. et al. Capturing dynamical correlations using implicit neural representations. *Nat. Commun.* **14**, 5822 (2023).
3. Ni, Y. et al. Physics-guided dual implicit neural representations for source separation. *Mach. Learn. Sci. Technol.* **6**, 045042 (2025).
4. Xie, Y. et al. Neural fields in visual computing and beyond. *Comput. Graph. Forum* **41**, 641–676 (2022).
5. Molaei, A. et al. Implicit neural representation in medical imaging: a comparative survey. In *Proc. IEEE/CVF International Conference on Computer Vision Workshops* 2373–2383 (IEEE, 2023).
6. Lu, Y., Jiang, K., Levine, J. A. & Berger, M. Compressive neural representations of volumetric scalar fields. *Comput. Graph. Forum* **40**, 135–146 (2021).
7. Yang, R. et al. Sharing massive biomedical data at magnitudes lower bandwidth using implicit neural function. *Proc. Natl Acad. Sci. USA* **121**, e2320870121 (2024).
8. Thayer, J., Mariani, V., Poitevin, F. & Wang, C. LCLS big data handling—how I learned to stop worrying and love the data deluge. *Synchrotron Radiat. News* **28**, 4–9 (2025).
9. Thayer, J. et al. Massive scale data analytics at LCLS-II. *EPJ Web Conf.* **295**, 13002 (2024).
10. Sobolev, E. et al. Data reduction activities at European XFEL: early results. *Front. Phys.* **12**, 1331329 (2024).
11. Essakine, A. et al. Where do we stand with implicit neural representations? A technical and performance survey. Preprint at https://arxiv.org/abs/2411.03688 (2025).
12. Strümpler, Y., Postels, J., Yang, R., Gool, L. V. & Tombari, F. Implicit neural representations for image compression. In *Proc. 17th European Conference on Computer Vision* 74–91 (Springer, 2022).
13. Müller, T., Evans, A., Schied, C. & Keller, A. Instant neural graphics primitives with a multiresolution hash encoding. *ACM Trans. Graph.* **41**, 102 (2022).
14. Sitzmann, V., Martel, J. N. P., Bergman, A. W., Lindell, D. B. & Wetzstein, G. Implicit neural representations with periodic activation functions. In *Proc. Advances in Neural Information Processing Systems* 7462–7473 (Curran Associates, 2020).
15. Mildenhall, B. et al. NeRF: representing scenes as neural radiance fields for view synthesis. In *Proc. 16th European Conference on Computer Vision* 405–421 (Springer, 2020).
16. Tancik, M. et al. Fourier features let networks learn high frequency functions in low dimensional domains. In *Proc. Advances in Neural Information Processing Systems* 7537–7547 (Curran Associates, 2020).
17. Cappello, F. et al. Lossy compression of scientific data: applications constrains and requirements. Preprint at https://arxiv.org/abs/2503.20031 (2025).
18. Cybenko, G. V. Approximation by superpositions of a sigmoidal function. *Math. Control Signals Syst.* **2**, 303–314 (1989).
19. Ni, Y. et al. Time-series X-ray diffraction from Cu_3Au. *Zenodo* https://doi.org/10.5281/zenodo.17344185 (2025).
20. Petsch, A. N. et al. High-energy spin waves in the spin-1 square-lattice antiferromagnet La_2NiO_4. *Phys. Rev. Res.* **5**, 033113 (2023).
21. Chen, H. et al. Testing the data framework for an AI algorithm in preparation for high data rate X-ray facilities. In *Proc. 4th Annual Workshop on Extreme-scale Experiment-in-the-Loop Computing* 1–9 (IEEE, 2022).
22. Favre-Nicolin, V., Leake, S. & Chushkin, Y. Free log-likelihood as an unbiased metric for coherent diffraction imaging. *Sci. Rep.* **10**, 2664 (2020).
23. Siemens-star ptychography demo dataset (ID01, ESRF, 2019); https://ftp.esrf.fr/pub/scisoft/PyNX/data/ptycho-siemens-star-id01.cxi
24. Galvez, R. et al. A machine-learning data set prepared from the NASA solar dynamics observatory mission. *Astrophys. J. Suppl. Ser.* **242**, 7 (2019).
25. Mallat, S. *A Wavelet Tour of Signal Processing: The Sparse Way* 3rd edn (Academic Press, 2008).
26. Shapiro, J. M. Embedded image coding using zerotrees of wavelet coefficients. *IEEE Trans. Signal Process.* **41**, 3445–3462 (1993).
27. Dupont, E., Goliński, A., Alizadeh, M., Teh, Y. W. & Doucet, A. COIN: compression with implicit neural representations. In *Proc. Neural Compression: From Information Theory to Applications (ICLR Workshop)* (OpenReview.net, 2021).
28. Dupont, E. et al. COIN++: neural compression across modalities. *Trans. Mach. Learn. Res.* **2022**, 1–29 (2022).
29. Jacot, A., Gabriel, F. & Hongler, C. Neural tangent kernel: convergence and generalization in neural networks. In *Proc. Advances in Neural Information Processing Systems* 8571–8580 (Curran Associates, 2018).
30. Lee, J. et al. Wide neural networks of any depth evolve as linear models under gradient descent. In *Proc. Advances in Neural Information Processing Systems* Vol. 32 8572–8583 (Curran Associates, 2019).
31. Rahaman, N. et al. On the spectral bias of neural networks. In *Proc. 36th International Conference on Machine Learning* 5301–5310 (PMLR, 2019).
32. Favre-Nicolin, V. Free log-likelihood as an unbiased metric for coherent diffraction imaging: figures and data. *Zenodo* https://doi.org/10.5281/zenodo.3451855 (2019).
33. Saragadam, V. et al. WIRE: wavelet implicit neural representations. In *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition* 18507–18516 (IEEE, 2023).
34. Liu, Z. et al. FINER: flexible spectral-bias tuning in implicit neural representation by variable-periodic activation functions. In *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition* 2713–2722 (IEEE, 2024).
35. Saragadam, V., Tan, J., Balakrishnan, G., Baraniuk, R. G. & Veeraraghavan, A. MINER: multiscale implicit neural representation. In *Proc. 17th European Conference on Computer Vision* 318–333 (Springer, 2022).
36. Martel, J. N. P. et al. ACORN: adaptive coordinate networks for neural scene representation. *ACM Trans. Graph.* **40**, 144 (2021).
37. Skodras, A., Christopoulos, C. & Ebrahimi, T. The JPEG 2000 still image compression standard. *IEEE Signal Process. Magazine* **18**, 36–58 (2001).

38. Ballé, J., Minnen, D., Singh, S., Hwang, S. J. & Johnston, N. Variational image compression with a scale hyperprior. In *Proc. 6th International Conference on Learning Representations* (OpenReview.net, 2018).
39. Liu, J. et al. High-performance effective scientific error-bounded lossy compression with auto-tuned multi-component interpolation. *Proc. ACM Manag. Data* **2**, 55 (2024).
40. Zhao, K. et al. Optimizing error-bounded lossy compression for scientific data by dynamic spline interpolation. In *Proc. 37th IEEE International Conference on Data Engineering* 1643–1654 (IEEE, 2021).
41. Liang, X. et al. SZ3: a modular framework for composing prediction-based error-bounded lossy compressors. *IEEE Trans. Big Data* **9**, 485–498 (2023).
42. Han, S., Mao, H. & Dally, W. J. Deep compression: compressing deep neural network with pruning, trained quantization and Huffman coding. In *Proc. 4th International Conference on Learning Representations* (2016).
43. Yang, C., Zhao, Y. & Wang, S. Deep image compression in the wavelet transform domain based on high frequency sub-band prediction. *IEEE Access* **7**, 52484–52497 (2019).
44. Hoefler, T., Alistarh, D., Ben-Nun, T., Dryden, N. & Peste, A. Sparsity in deep learning: pruning and growth for efficient inference and training in neural networks. *J. Mach. Learn. Res.* **22**, 241 (2021).
45. Han, J. & Yang, F. DCINR: a divide-and-conquer implicit neural representation for compressing time-varying volumetric data in hours. *IEEE Trans. Vis. Comput. Graph.* **31**, 8116–8128 (2025).
46. Yang, R., Xiao, T., Cheng, Y., Suo, J. & Dai, Q. TINC: tree-structured implicit neural compression. In *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition* 18517–18526 (IEEE, 2023).
47. Cai, Z., Zhu, H., Shen, Q., Wang, X. & Cao, X. Batch normalization alleviates the spectral bias in coordinate networks. In *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition* 25160–25171 (IEEE, 2024).
48. Shi, K., Chen, H., Zhang, L. & Gu, S. Inductive gradient adjustment for spectral bias in implicit neural representations. *Proc. Mach. Learn. Res.* **267**, 54864–54891 (2025).
49. McGinnis, J. et al. Optimizing rank for high-fidelity implicit neural representations. Preprint at https://arxiv.org/abs/2512.14366 (2025).
50. Zheng, J., Saratchandran, H. & Lucey, S. The inlet rank collapse in implicit neural representations: diagnosis and unified remedy. Preprint at https://arxiv.org/abs/2602.01526 (2026).
51. Taubman, D. S. & Marcellin, M. W. *JPEG2000: Image Compression Fundamentals, Standards and Practice* 1st edn (Springer, 2002).
52. Coifman, R. R. & Wickerhauser, M. V. Entropy-based algorithms for best basis selection. *IEEE Trans. Inf. Theory* **38**, 713–718 (1992).
53. Huang, J. J. & Dragotti, P. L. WINNet: wavelet-inspired invertible network for image denoising. *IEEE Trans. Image Process.* **31**, 4377–4392 (2022).
54. Liu, Z. et al. KAN: Kolmogorov–Arnold networks. In *Proc. 13th International Conference on Learning Representations* (OpenReview.net, 2025).
55. Hui, K. H., Li, R., Hu, J. & Fu, C.-W. Neural wavelet-domain diffusion for 3D shape generation. In *Proc. SIGGRAPH Asia 2022 Conference Papers* 1–9 (ACM, 2022).
56. You, T., Kim, M., Kim, J. & Han, B. Generative neural fields by mixtures of neural implicit functions. In *Proc. Advances in Neural Information Processing Systems* Vol. 36 20352–20370 (Curran Associates, 2023).
57. Serrano, D., Szymkowiak, J. & Musialski, P. HOSC: a periodic activation function for preserving sharp features in implicit neural representations. Preprint at https://arxiv.org/abs/2401.10967 (2024).
58. Kazerouni, A., Azad, R., Hosseini, A., Merhof, D. & Bagci, U. INCODE: implicit neural conditioning with prior knowledge embeddings. In *Proc. IEEE/CVF Winter Conference on Applications of Computer Vision* 1298–1307 (IEEE, 2024).
59. Shi, K., Zhou, X. & Gu, S. Improved implicit neural representation with Fourier reparameterized training. In *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition* 25985–25994 (IEEE, 2024).
60. Fathony, R., Sahu, A. K., Willmott, D. & Kolter, J. Z. Multiplicative filter networks. In *Proc. International Conference on Learning Representations* (OpenReview.net, 2021).
61. Yu, C., Luo, Y., Ye, K., Zhao, X. & Meng, D. Cross-frequency implicit neural representation with self-evolving parameters. Preprint at https://arxiv.org/abs/2504.10929 (2025).
62. Chitturi, S. R. et al. Capturing dynamical correlations using implicit neural representations. *Zenodo* https://doi.org/10.5281/zenodo.8267499 (2023).
63. PyNX public data repository: ptycho-siemens-star-id01.cxi. *PyNX* http://ftp.esrf.fr/pub/scisoft/PyNX/data/ptycho-siemens-star-id01.cxi (accessed 4 March 2026).
64. Kodak. Kodak lossless true color image suite. *Kodak* https://r0k.us/graphics/kodak/ (accessed 2 March 2026).
65. Ni, Y. et al. Multi-resolution enhancement for full spectrum neural representations. *Code Ocean* 10.24433/CO.8219927.v1 (2026).
66. torchao. TorchAO: PyTorch-native training-to-serving model optimization. *GitHub* https://github.com/pytorch/ao (2024).

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Author contributions

Y.N. conceived the WIEN-INR methodology, developed the machine learning pipeline and software implementation, and performed the computational experiments and evaluations. Z.C. contributed to code development and provided input on numerical evaluations and benchmarking. S.X. developed the mathematical theory and proofs and wrote the theoretical section of the paper. R.P. contributed the analysis tools, provided and prepared the experimental data, and provided input on numerical experiments. C.P. contributed to the numerical experiment design and paper organization. C.H.Y. contributed to the initial concept of applying neural representations to scientific data compression as well as the early benchmarking implementation. J.B.T. supervised the project and advised on the scientific use cases and broader applicability. J.J.T. supervised and coordinated the project, advised on dataset selection and scientific use cases, and contributed to paper writing. Y.N. wrote the paper with input from all authors. All authors discussed the results and implications.

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Competing interests

The authors declare no competing interests.

Additional information

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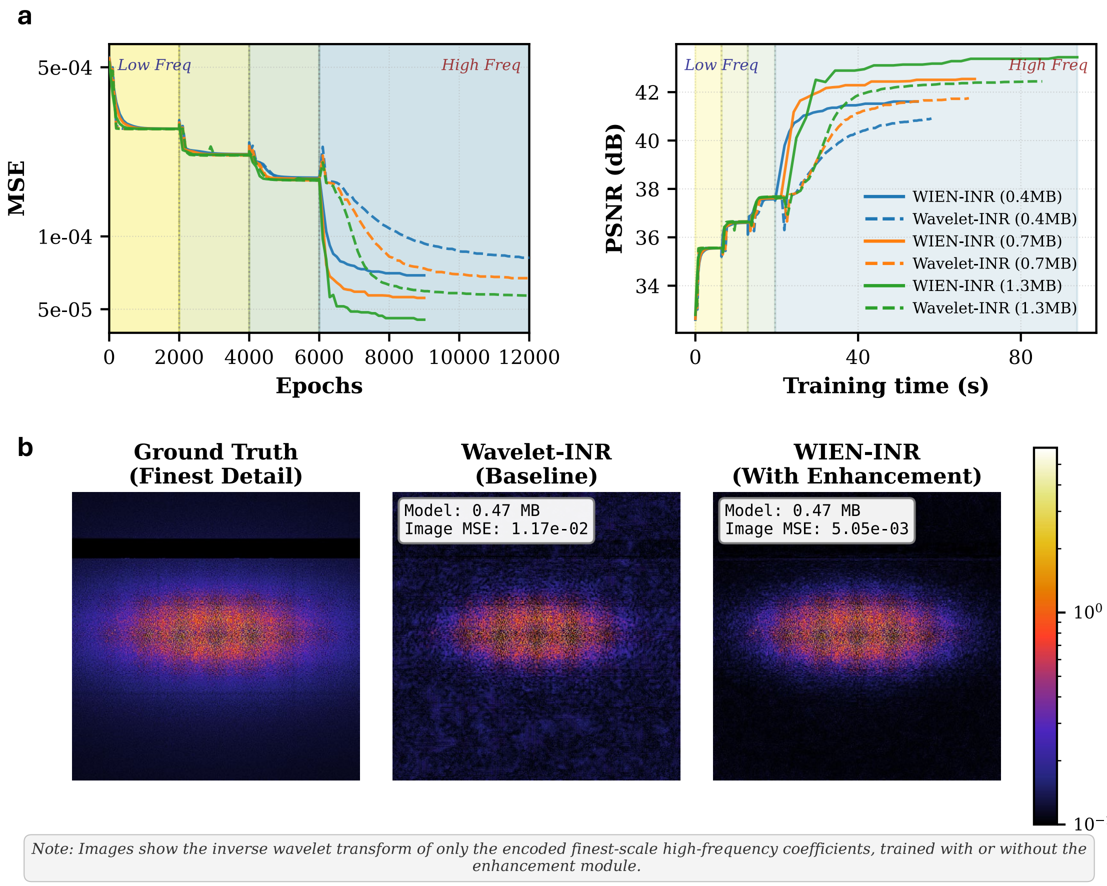

Extended Data Fig. 1 | Effect of the enhancement module on convergence and finest-scale fidelity. a, Training curves on Cu3Au[19] using models with matched sizes (colors), comparing WIEN-INR with enhancement (solid) to the WAVELET-INR baseline without enhancement (dashed). The enhancement module yields faster convergence in the finest band, with smaller loss jumps when transitioning across wavelet scales and lower final loss. This stabilization is consistent with (i) improved initialization of the finest band from coarser-scale predictions and (ii) a learnable overlapping kernel that refines coarser outputs, making fitting the most difficult details easier. **b**, Reconstruction of the finest-band detail transformed back to image space via inverse wavelet transform. The enhancement module reduces error, visual artifacts and better preserves speckle contrast relative to the baseline.

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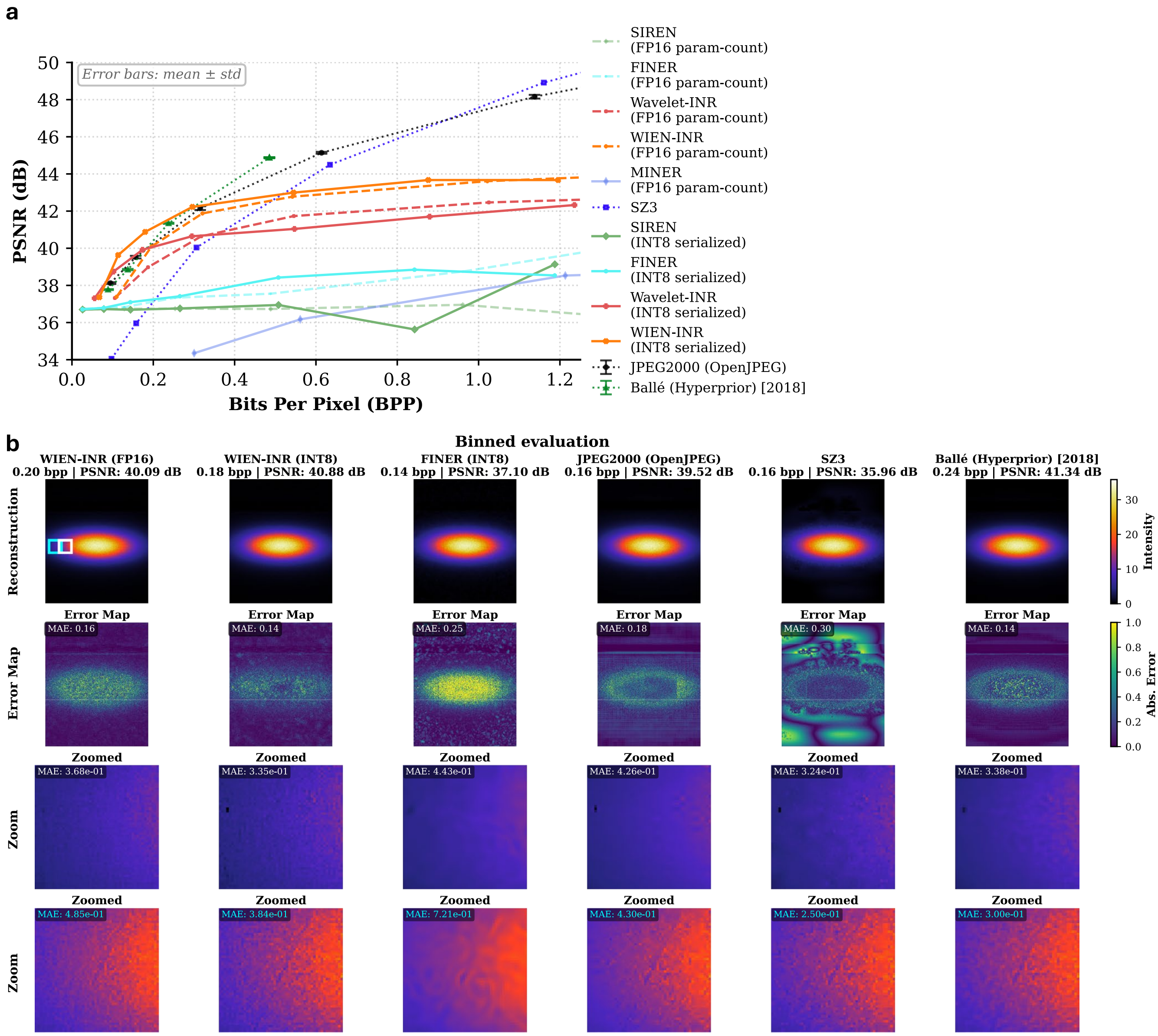

Extended Data Fig. 2 | Comparison to compressor baselines on the X-ray scattering Cu3Au sample. a, Rate-distortion (PSNR versus bits per pixel (bpp)) for INR methods, JPEG-2000 (ref. 37), the scientific compressor SZ3 [39–41], and autoencoder based learned compressor BMS (Ballé (Hyperprior) [2018])[38]. JPEG-2000 and the BMS are applied slice-by-slice; we report mean ± std with n = 36 across slices. For INRs, we report both FP16 parameter count and an INT8 weight-only serialized variant (using TorchAO int8 weight only [66]). While JPEG-2000 and the learned image codec can achieve PSNR comparable to WIEN-INR at low bpp regime, they exhibit higher MAE and reduced speckle contrast in zoomed regions. WIEN-INR outperforms SZ3 at low bpp, while SZ3 remains highly competitive at higher bitrates, consistent with its maturity in near-lossless scientific compression. **b**, Binned reconstructions (computed by averaging the decoded 3D volume along the time axis) are shown for all methods at approximately 0.2 bpp, with zoomed-in ROIs. Reported PSNR for the JPEG-2000 and BMS are computed on the full volume. SZ3 remains robust in the signal areas while some artifacts are observed at the background. WIEN-INR and SZ3 preserve fine-scale speckle structure and yields lower local error in both strong- and weak-signal regions.

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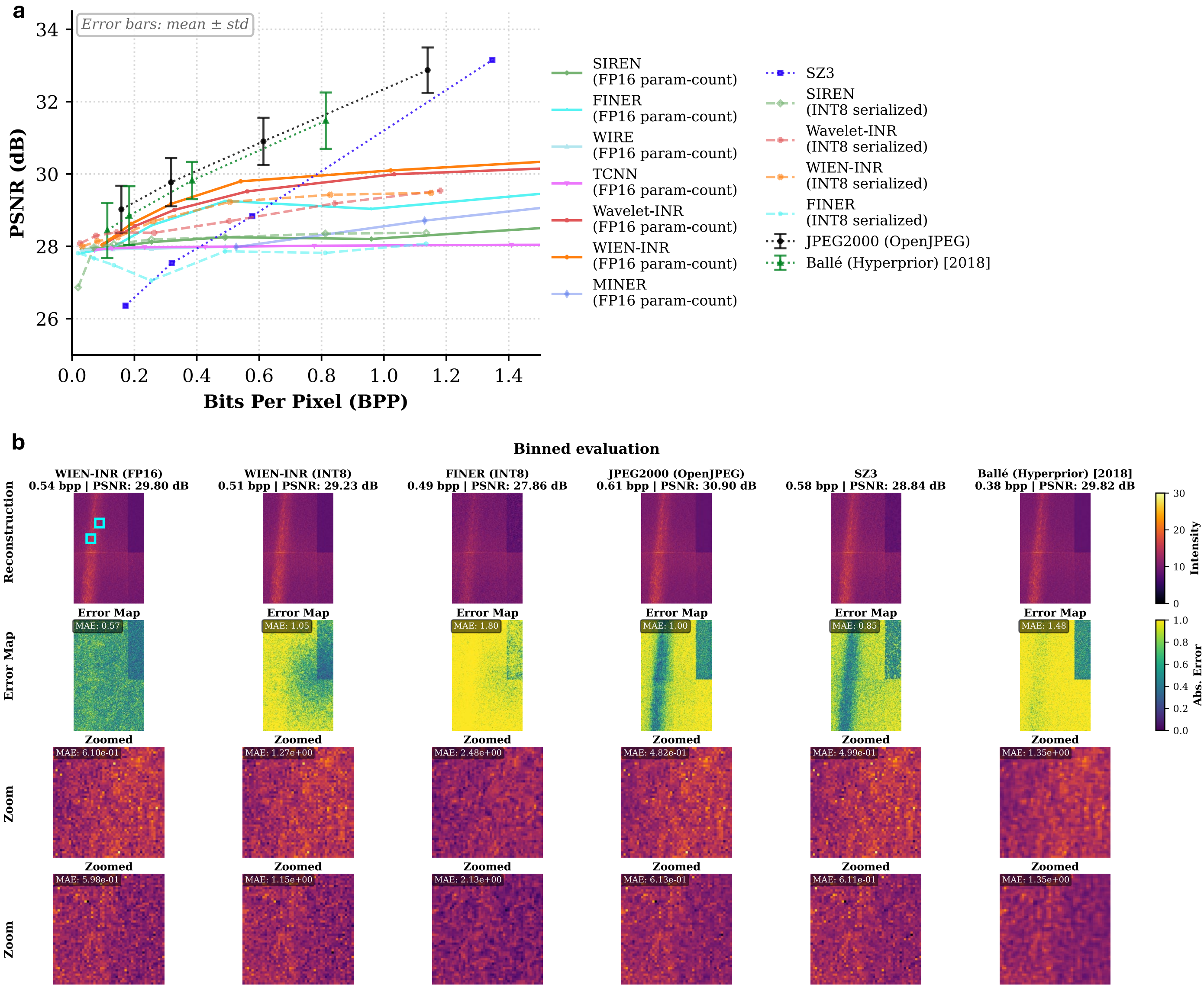
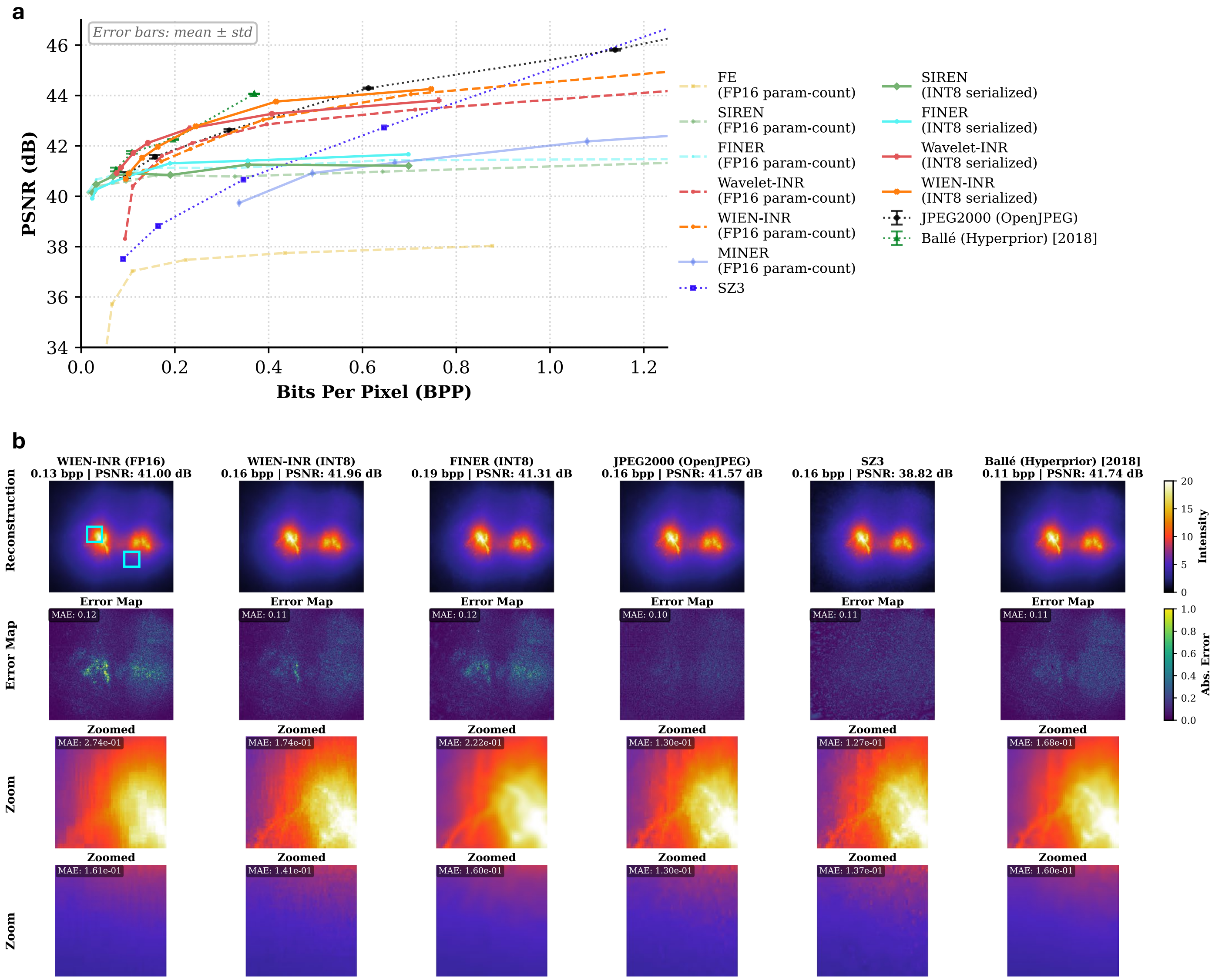


Extended Data Fig. 3 | Comparison to compressor baselines on the XFEL wide-angle X-ray scattering NiPS3 sample. Rate-distortion curves (panel **a**) and reconstructions (panel **b**) showing time-averaged images computed from n=100 two-dimensional time frames, with error maps and zoomed-in ROIs highlighting speckle structure. At matched bpp, JPEG2000 achieves the highest mean PSNR but exhibits 2 × higher MAE than WIEN-INR (FP16) with visible artifacts in the zoom-in plots. The VAE-based codec (Ballé/CompressAI) shows 3 × higher MAE and near-complete loss of speckle contrast despite competitive PSNR. WIEN-INR outperforms SZ3 at low bpp, while SZ3 remains strong at higher rates consistent with its near-lossless design. These results demonstrate that quality metrics optimized for natural images are unsuitable for this speckle-dominated scattering data. INR-based compressors exhibit sensitivity to INT8 weight-only quantization (FP32 → INT8)[66].



Extended Data Fig. 4 | Comparison to compressor baselines on the coherent X-ray diffraction Fe(Te,Se)(300K) sample. We plot the PSNR-bpp curve in panel **a** and reconstructions in **b**. We report time-averaged images computed from n=20 two-dimensional time frames, with error maps and zoomed-in ROIs. Consistent with the other two cases, we observe that PSNR alone does not reliably reflect visual/physics-related fidelity for these scientific measurements: JPEG2000 and the VAE-based codec (Ballé/CompressAI), which are optimized for natural-image statistics, can suppress fine-scale contrast and oversmooth the physics related features in the ROI zooms, where weak speckles and diffuse scattering may be treated as noise by their quality controls. Across rates, WIEN-INR outperforms all INR baselines and achieves higher PSNR than SZ3 in the low-bpp regime, while SZ3 remains competitive at higher rates. In this dataset, the INT8 weight-only variant performs comparably to (or better than) FP16.

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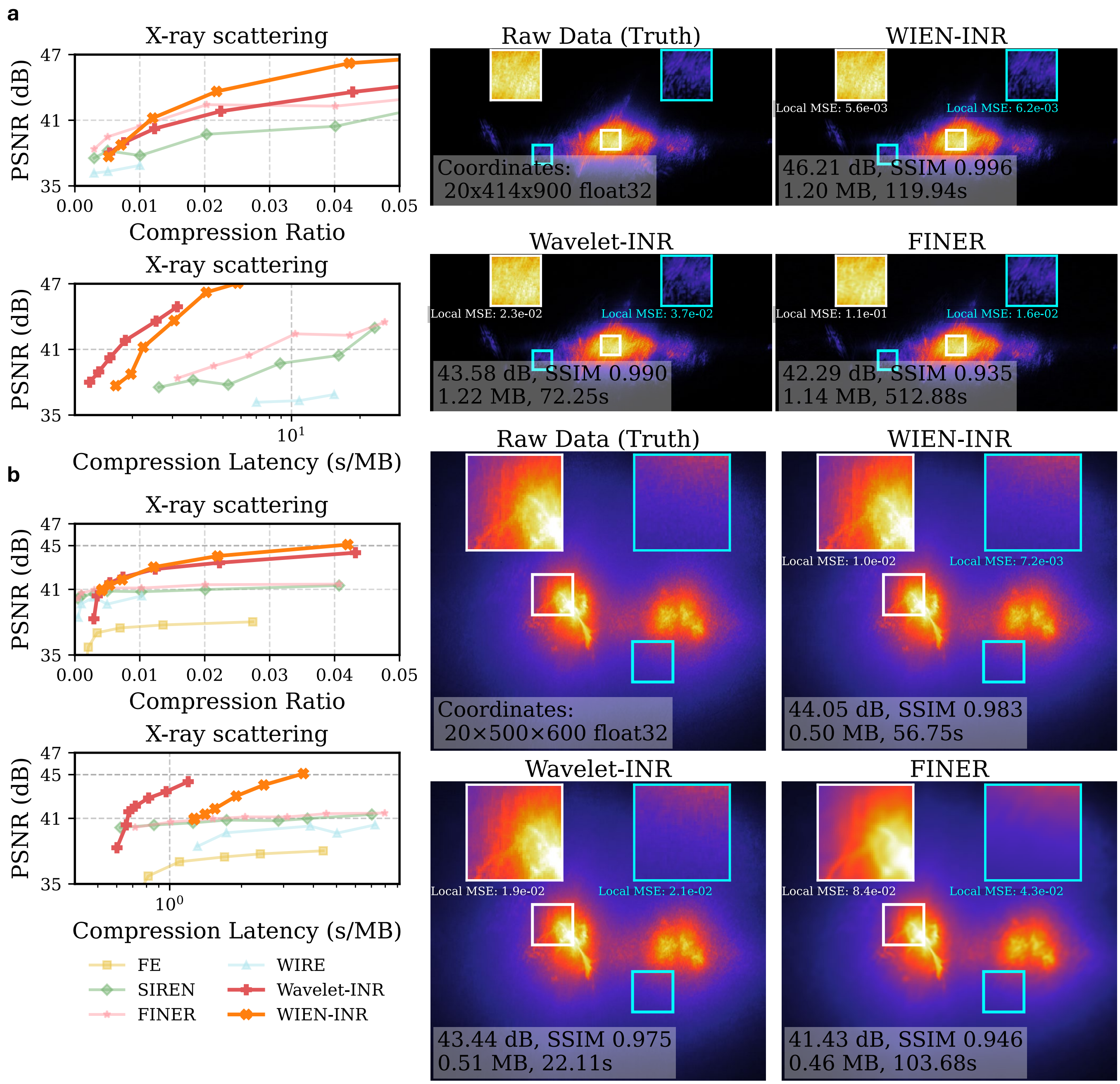
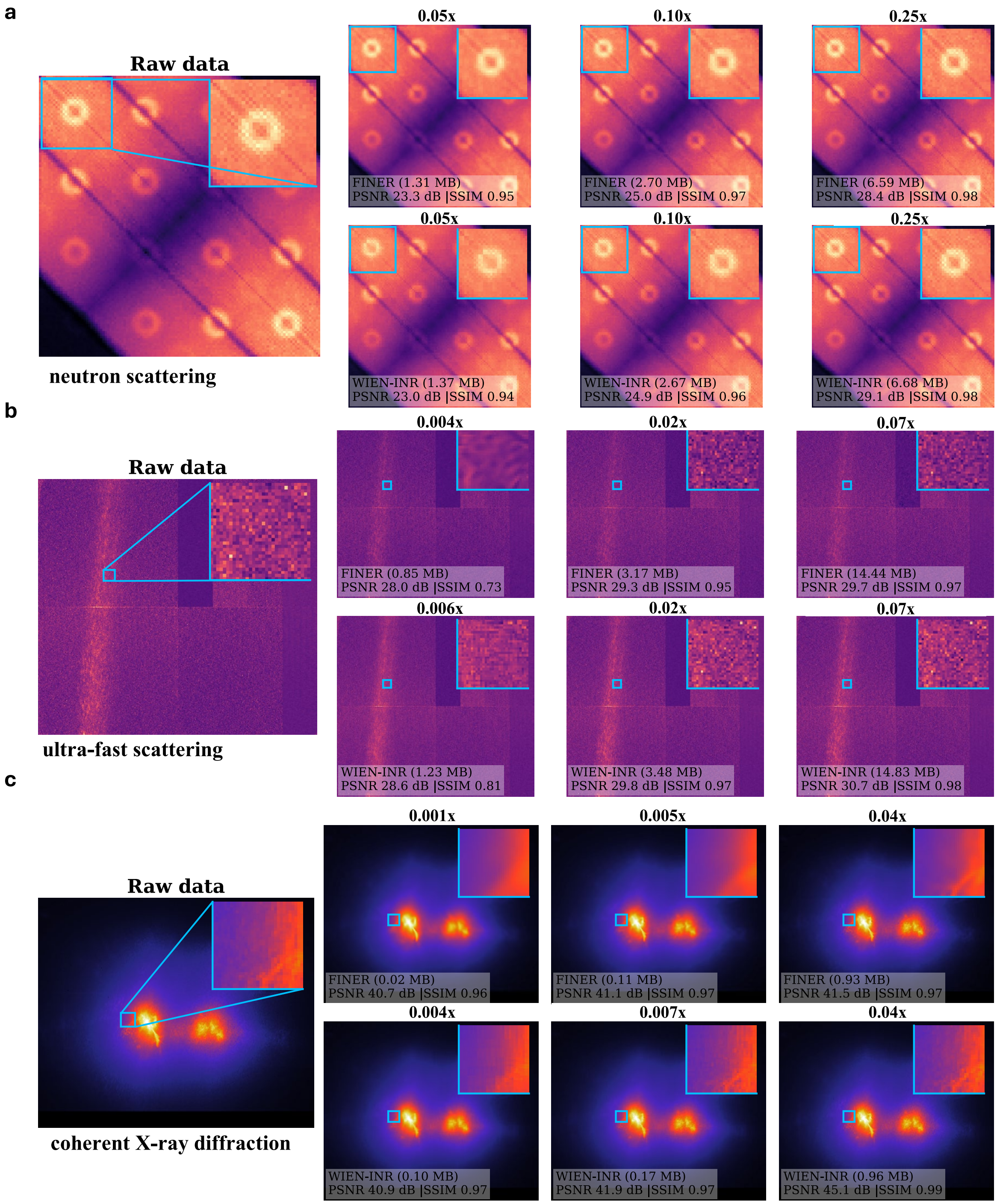
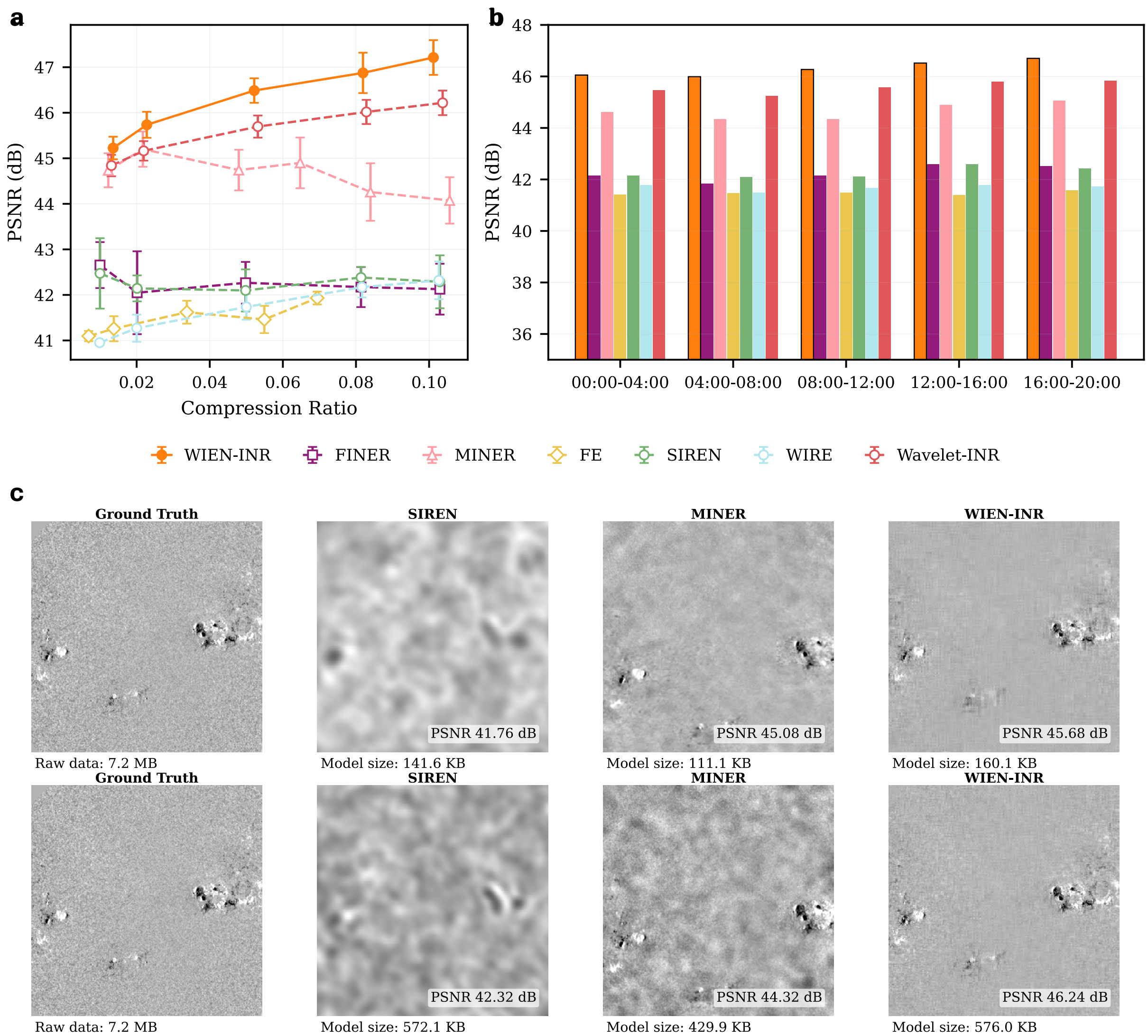


Extended Data Fig. 5 | Extended examples on two additional time-series data. Performance comparison on time-series dataset from coherent X-ray diffraction on **a**, Fe(Te,Se)(120K) and **b**, Fe(Te,Se)(300K) samples, collected at Beamline 11-ID at the National Synchrotron Light Source II at Brookhaven National Laboratory. Beyond a low parameter budget (for example, after 100× compression for a), WIEN-INR achieves consistent improved performance in terms of global PSNR compared to the baselines. Local loss metrics and zoomed-in views also provide supplementary analyses to characterize where fidelity is lost beyond the global volume-level metrics. The right panels demonstrate the reconstruction of the data aggregated along the time-axis, with inset zooms to ROIs of Bragg peaks and diffuse scattering regions. WIEN-INR preserves speckle contrast and structure without smoothing artifacts or unphysical textures observed in the FINER baseline. Inset texts additionally report local ROI MSE, demonstrating WIEN-INR achieves ~ 2-7× lower error to the best performing baseline (FINER).



Extended Data Fig. 6 | Additional decoded volumes under small model-size budgets. Decoded volumes of the encoded **a**, neutron scattering[20], **b**, ultra-fast X-ray scattering[21], and **c**, coherent X-ray diffraction[19] datasets for various compressive INR model sizes, with compression ratios indicated in the column titles. This figure complements Fig. 4 by showing additional examples from the low-parameter/high-rate region of the rate-distortion curves, where the achieved PSNR has not yet saturated and methods may exhibit visible reconstruction artifacts. The purpose of this comparison is not to assess overall performance from these examples alone, but to illustrate the types of artifacts that appear under constrained model-size budgets and how they vary across methods and datasets. We decode the full 3D/4D volume and then sum along either the time axis or the (l, ω) axis of the neutron data for visual inspection. Inset text reports model sizes and the achieved PSNR/SSIM metrics.



Extended Data Fig. 7 | Extended examples on solar magnetograms from SDO/HMI. Evaluation of WIEN-INR on a 20-hour time series of SDO/HMI magnetogram imagery from the curated SDO machine-learning dataset of Galvez et al.[24]. **a**, Rate-distortion (PSNR versus compression ratio) for models trained on sequential 4-hour segments, with mean ± std across five segments. **b**, Per-segment PSNR for each model, showing consistent performance across the 20-hour period. **c**, Representative reconstructions at time 00:00 for small (top row) and large (bottom row) model sizes, displayed in log scale. WIEN-INR consistently outperforms INR baselines while SIREN completely degenerates under tight parameter budgets, while MINER produces obvious patchy artifacts.

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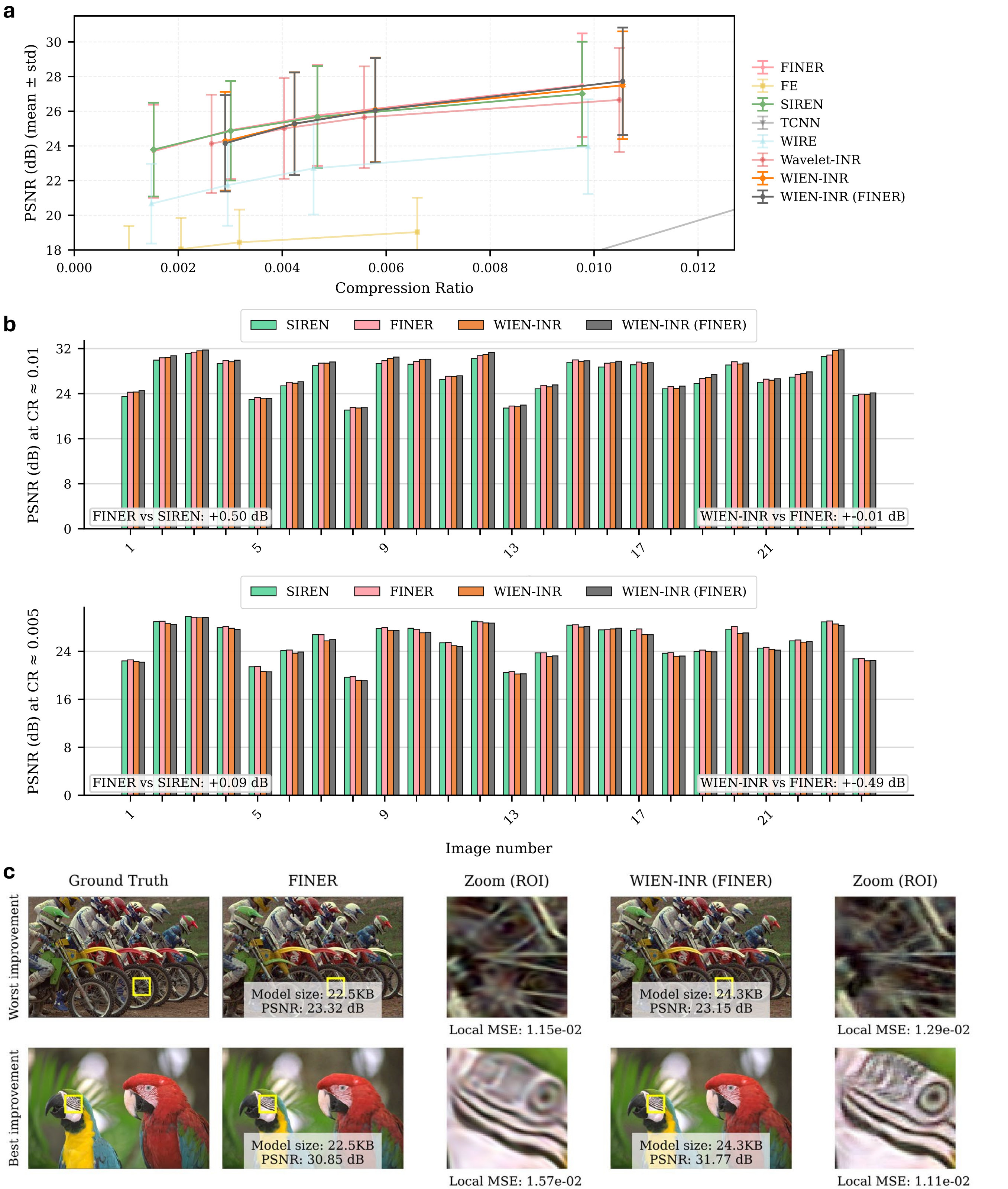

Extended Data Fig. 8 | Performance on natural images. Performance on the Kodak dataset[64] consists of 24 natural photography images. WIEN-INR (FINER) uses FINER as the backbone at each scale. **a**, Mean ± std PSNR across n = 24 images, each fitted separately. At very low compression rates, multi-scale overhead reduces performance relative to global methods (SIREN, FINER); WIEN-INR variants outperform baselines at larger network sizes. **b**, Per-image PSNR at compression rates 0.005 and 0.01. We observe that at low compression rates (0.005), since multi-scale decomposition methods distribute parameters across multiple frequency bands, meaning smoother bands—which dominate natural images—receive smaller sub-networks, potentially disadvantaging overall PSNR compared to single-network approaches. And enhancement introduces unnecessary computational and encoding overhead. At higher compression rates (0.01), WIEN-INR (FINER) shows consistent gains. **c**, We plot the two images where WIEN-INR (FINER) multi-scale method performs worst and best over FINER. A key finding is that natural images with sharp boundaries and cartoon-like features (such as the stripes around the bird's eye) benefit from WIEN-INR's multi-resolution decomposition and enhancement.

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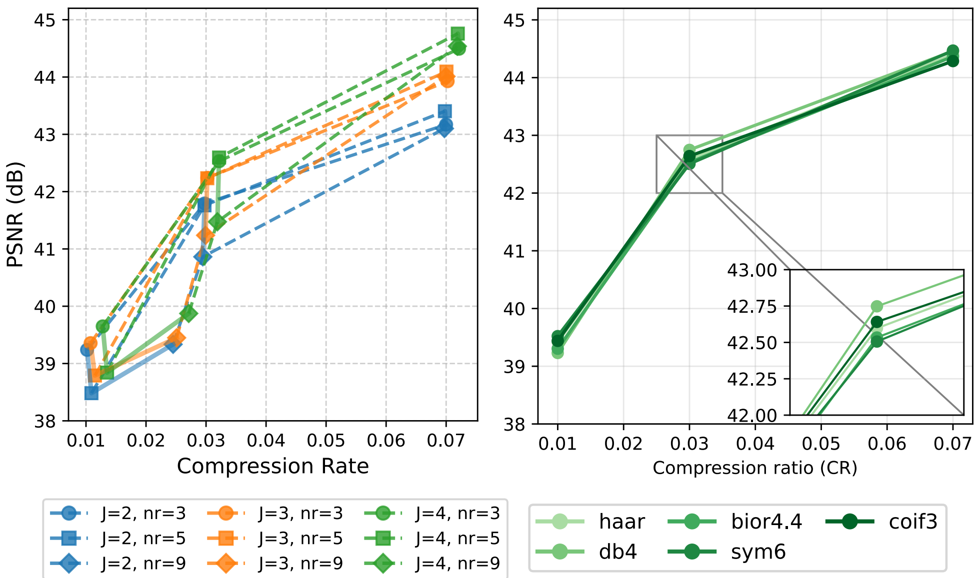

Extended Data Fig. 9 | Robustness to wavelet choices. (Left) Use a fixed wavelet type with varying depth J and r to encode the Cu3Au sample[19] using WIEN-INR. We plot the rate-distortion curves using Haar wavelet with different multi-resolution depths $J \in \{2,3,4\}$ and neighborhood sizes $n_r = r \in \{3,5,9\}$. We plot the compression rate (network size over raw size) versus the achieved accuracy (PSNR (dB)). Larger J and n_r induce larger networks, but potentially higher representation power. Colors indicate different J, markers indicate different r; curves with the same r are connected by dashed lines, and curves with the same J by solid lines. (Right) Study of using a fixed depth J and r but varying wavelet types. Rate-distortion curves for J = 4 and r = 3 are plotted, where our network is trained to encode DWT coefficients computed with five different wavelet bases: Haar, Biorthogonal 4.4 (bior4.4), Coiflet 3 (coif3), Daubechies 4 (db4), and Symlet 6 (sym6), indicated by different shades of green colors. We find J = 4 provides the best balance of frequency separation, while performance remains stable across wavelet choices, with Haar at J = 4, nr = 3 yielding strong results across all benchmark datasets.

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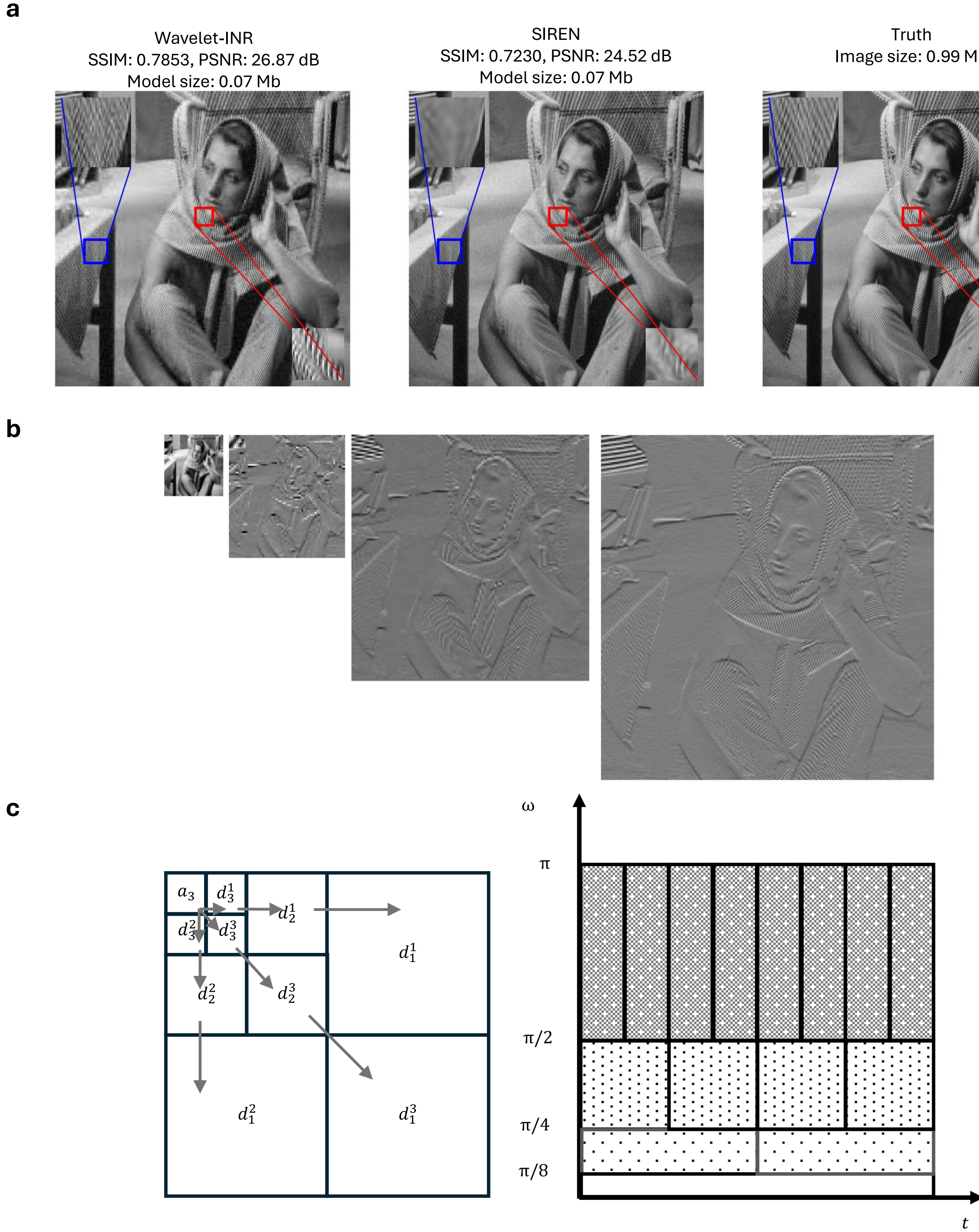

Extended Data Fig. 10 | Multi-resolution wavelet decomposition with implicit neural representations. a. Neural representation of the Barbara image using different INR approaches. Left: Separate INRs for each DWT sub-band, which is the same as our baseline WAVELET-INR. Center: A single network representing all frequency bands jointly. Right: Ground truth image. With the same total number of network parameters, the left decomposition assigns more representational capacity to high-frequency bands, yielding a representation with enhanced fine details as seen in the enlarged textures. **b**, Example wavelet coefficients of the Barbara image, showing approximation coefficients $\mathbf{a}_3$ and detail coefficients $\boldsymbol{d}_3^1$, $\boldsymbol{d}_2^1$, and $\boldsymbol{d}_1^1$ (from left to right) using the haar wavelet. **c**, **left**, Schematic of a two-dimensional wavelet transform, where the lowest-frequency approximation sub-band $\mathbf{a}_3$ occupies the top-left quadrant, and progressively higher-frequency detail sub-bands appear toward the bottom right. Within each scale j, the three detail sub-bands ($\boldsymbol{d}_j^1$, $\boldsymbol{d}_j^2$, $\boldsymbol{d}_j^3$) capture horizontal, vertical, and diagonal orientations, respectively. **c**, **right**, Corresponding tiling of the time-frequency plane. Each successive scale approximately halves the bandwidth in the frequency domain, with scale j responding primarily to the octave $[\pi/2^j, \pi/2^{j-1}]$, subject to overlap due to the non-ideal frequency localization of practical wavelets.

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Data collection	Our data is sourced from controlled physical experiments conducted at multiple facilities. A more detailed description of the experimental procedures and data collection methodologies can be found in the Supplementary Information (SI Data).
Data analysis	Data analysis and model evaluation were carried out using custom Python 3.12.4, PyTorch 2.4.0 with CUDA 12.4, NumPy 2.1.0, Matplotlib 3.9.2, scikit-image 0.25.2, pytorch-wavelets 1.3.0 and PyWavelets 1.7.0. The custom code used to implement WIEN-INR, train and evaluate the WIEN-INR methods, and generate the main analysis scripts is deposited in a Code Ocean compute capsule with a citable DOI: https://doi.org/10.24433/CO.8219927.v1. The development repository is available at https://github.com/YN754/INC4Sci.git.

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The primary Cu3Au datasets that support the main claims and figures of this study have been deposited in Zenodo https://doi.org/10.5281/zenodo.17344185. Neutron scattering data are available in Zenodo https://doi.org/10.5281/zenodo.8267499. Additional datasets used in this study are publicly available, they include: coherent diffraction imaging of the ESRF logo; Siemens-star ptychography data from the PyNX ESRF public folder; solar magnetogram data from the NASA Solar Dynamics Observatory Helioseismic and Magnetic Imager (HMI), analyzing the first 20 hours from 25 February 2014; and the Kodak dataset from the Kodak True Color Image Suite. The coherent X-ray scattering from Fe(Te,Se) and ultra-fast X-ray scattering data from single-crystal NiPS3 were collected under beamtime agreements that restrict public sharing. These restricted datasets are available from the corresponding author, Joshua J. Turner (joshuat@slac.stanford.edu), upon reasonable request and subject to approval under the applicable beamtime and facility data-sharing policies; requests will be considered within four weeks.

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Randomization	This study does not involve life sciences study design.
Blinding	This study does not involve life sciences study design.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Plants

Seed stocks	This study did not involve plants.
Novel plant genotypes	This study did not involve plants.
Authentication	This study did not involve plants.